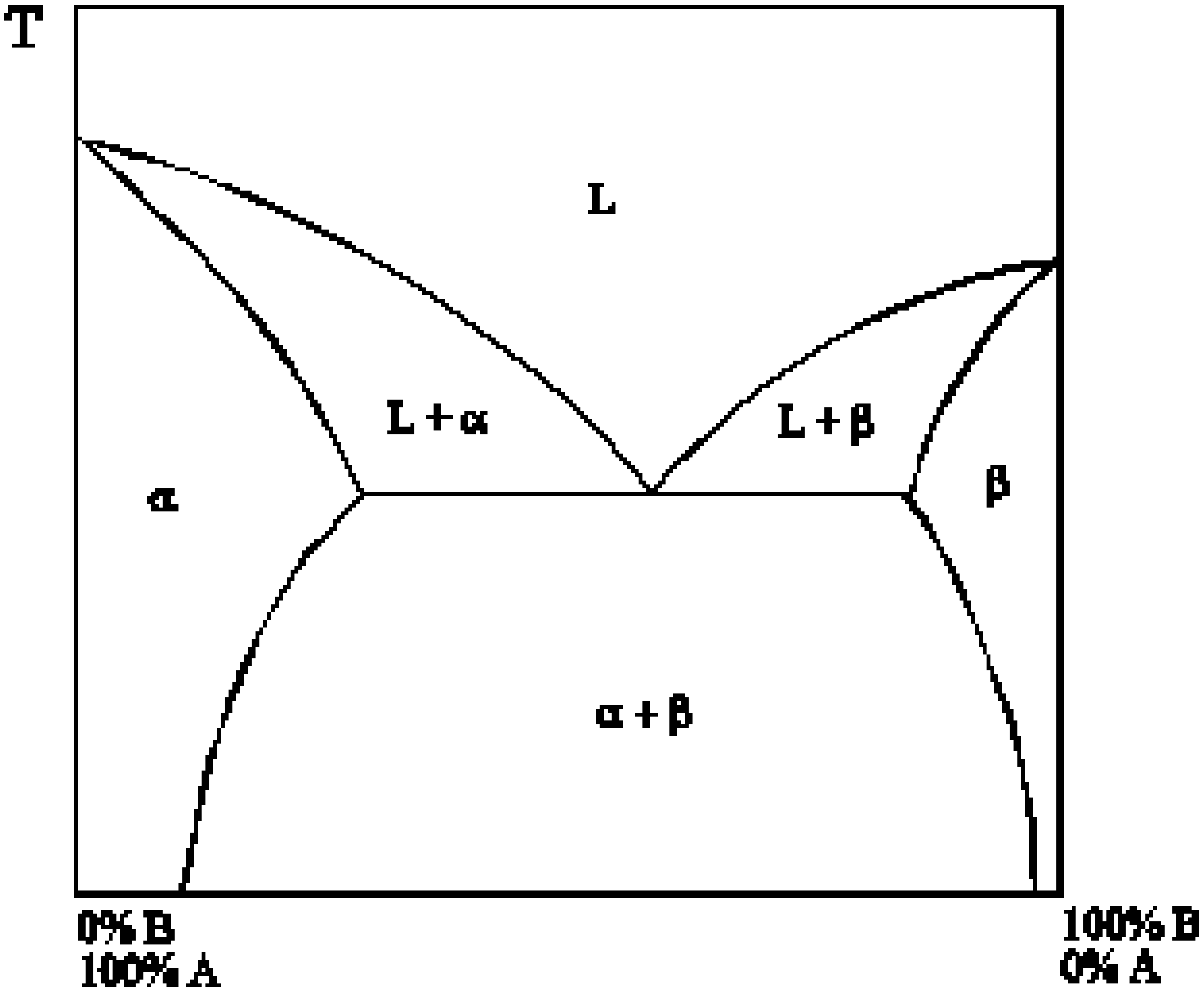
STM Exercise 7

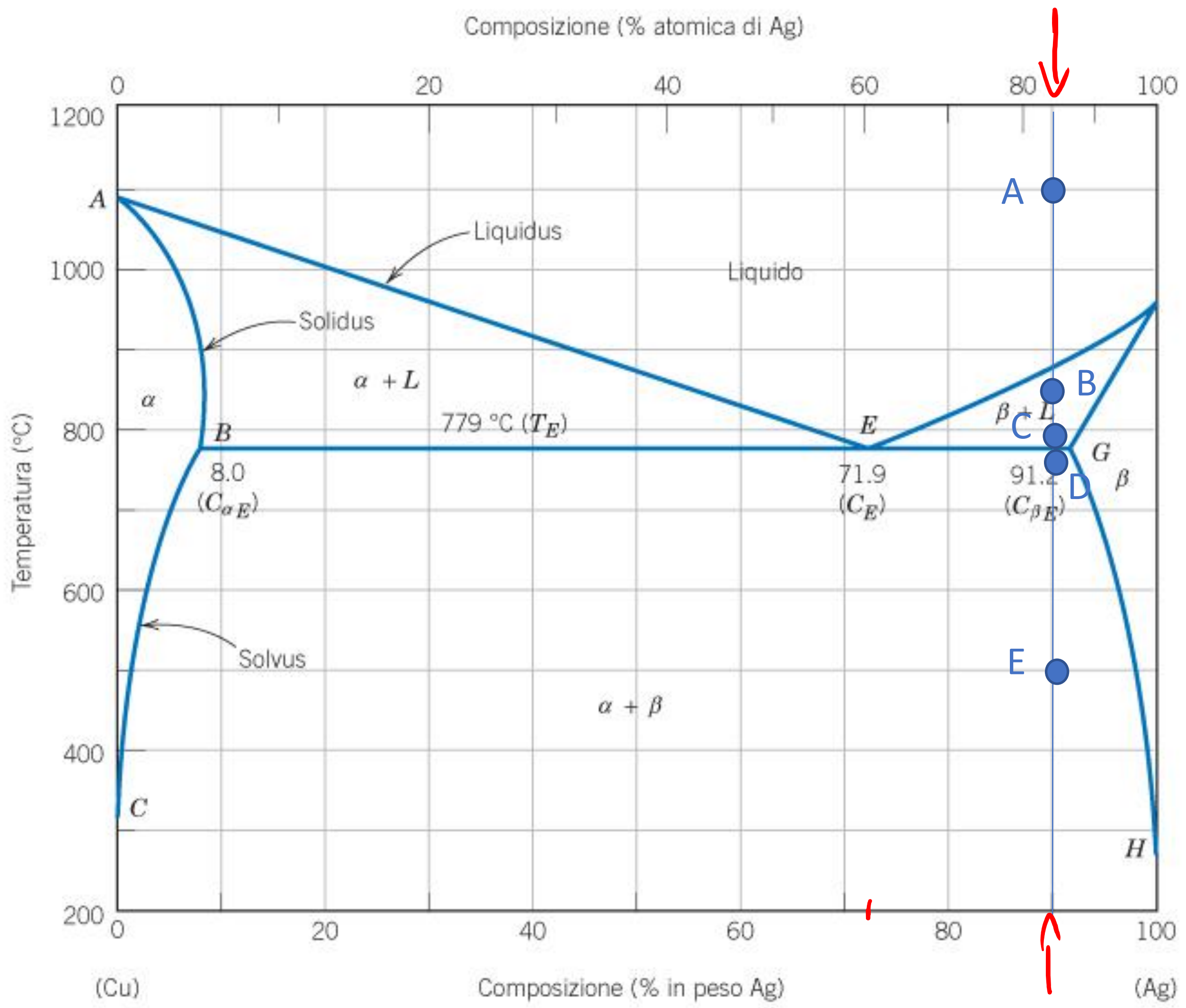
Phase Diagrams

(partial solubility in the solid state)

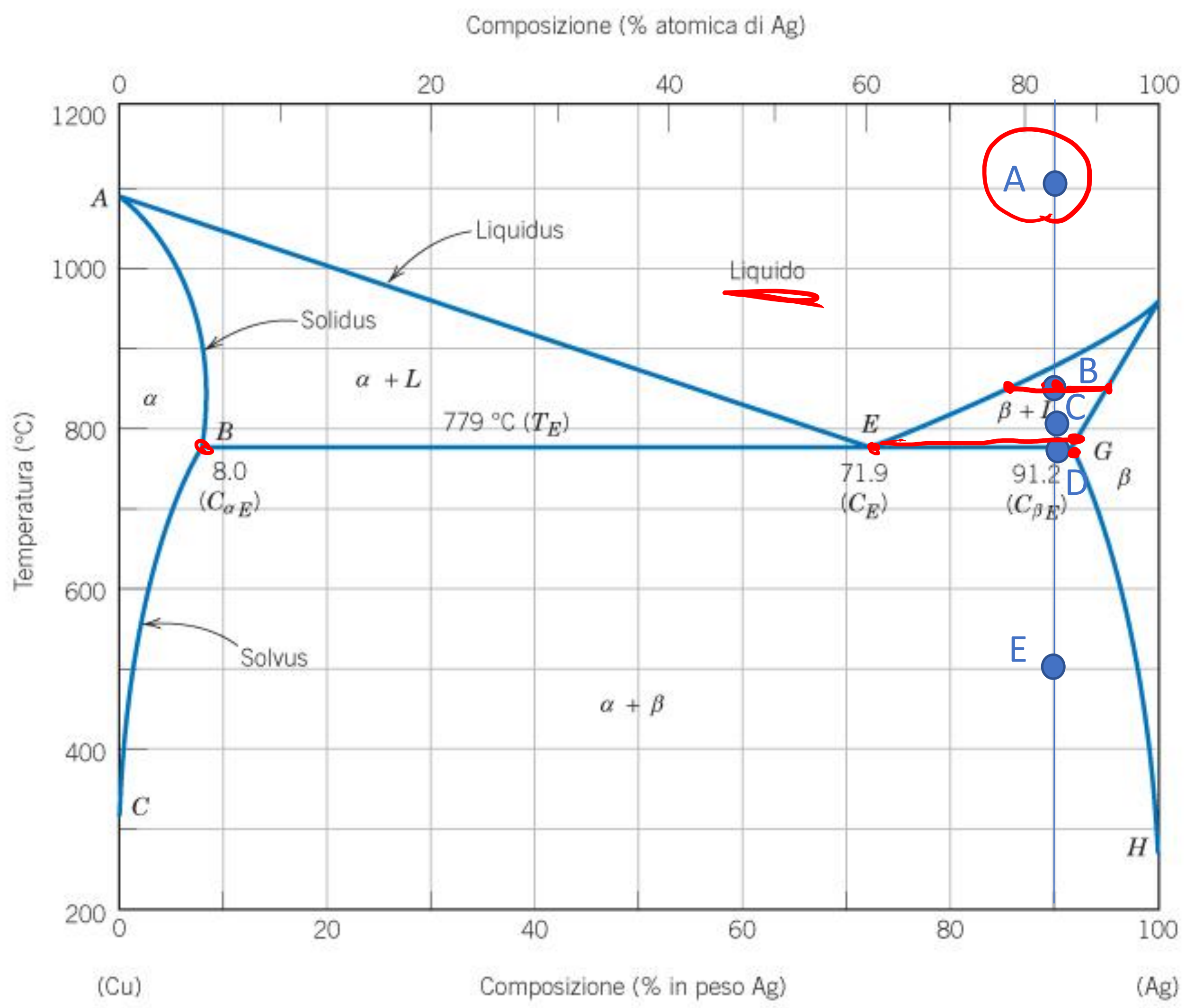
Complete solubility in the liquid state, partial solubility in the solid state



Exercise 1: determine the phase present, their composition and relative amounts in the points A, B, C (at the beginning of the eutectic transformation), D (at the end of the eutectic transformation) and E indicated in the phase diagram. Schematize the microstructure in each point.



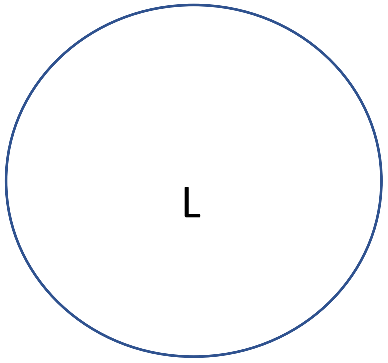
C_0 $\begin{cases} 90\%\ Ag \\ 10\%\ Cu \end{cases}$

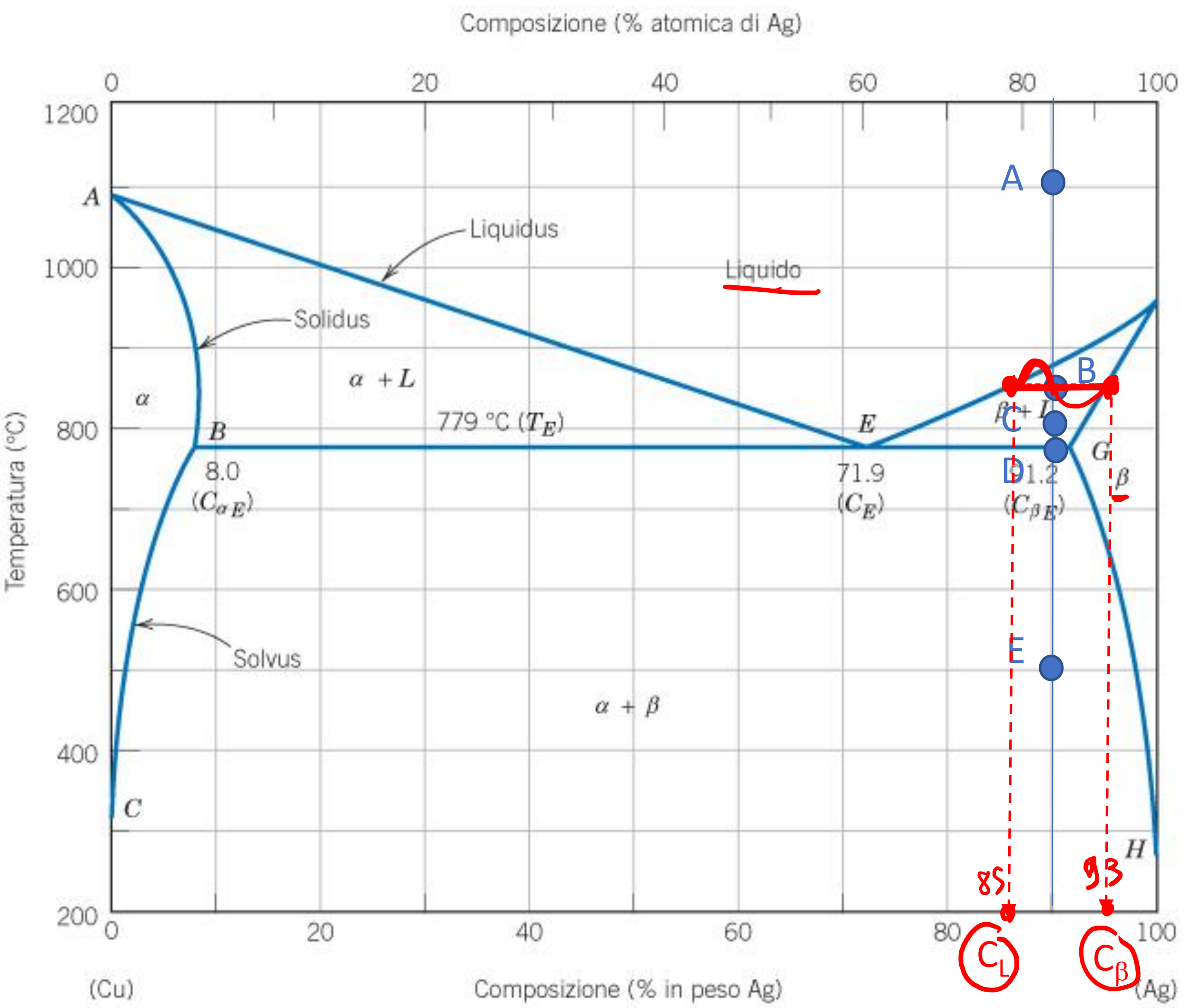


Point A:
Phases: 1 phase liquid

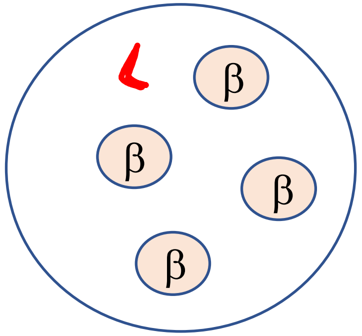
Composition: 90%Ag, 10%Cu

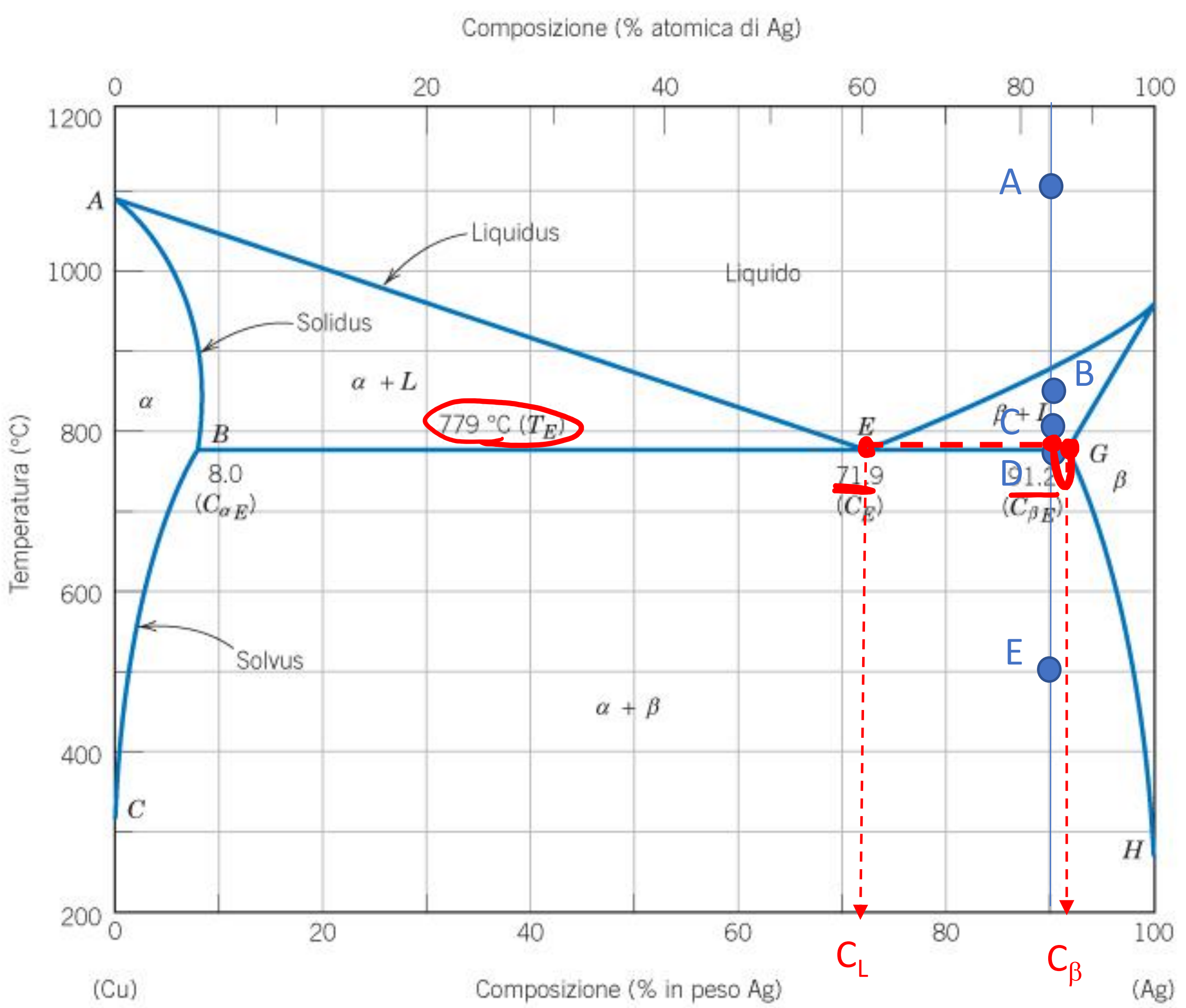
Relative amount: 100% liquid phase



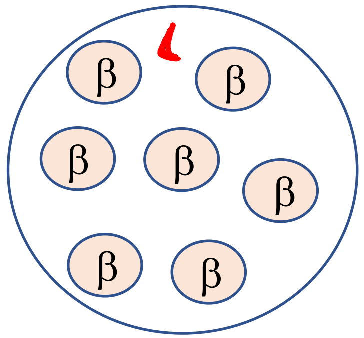


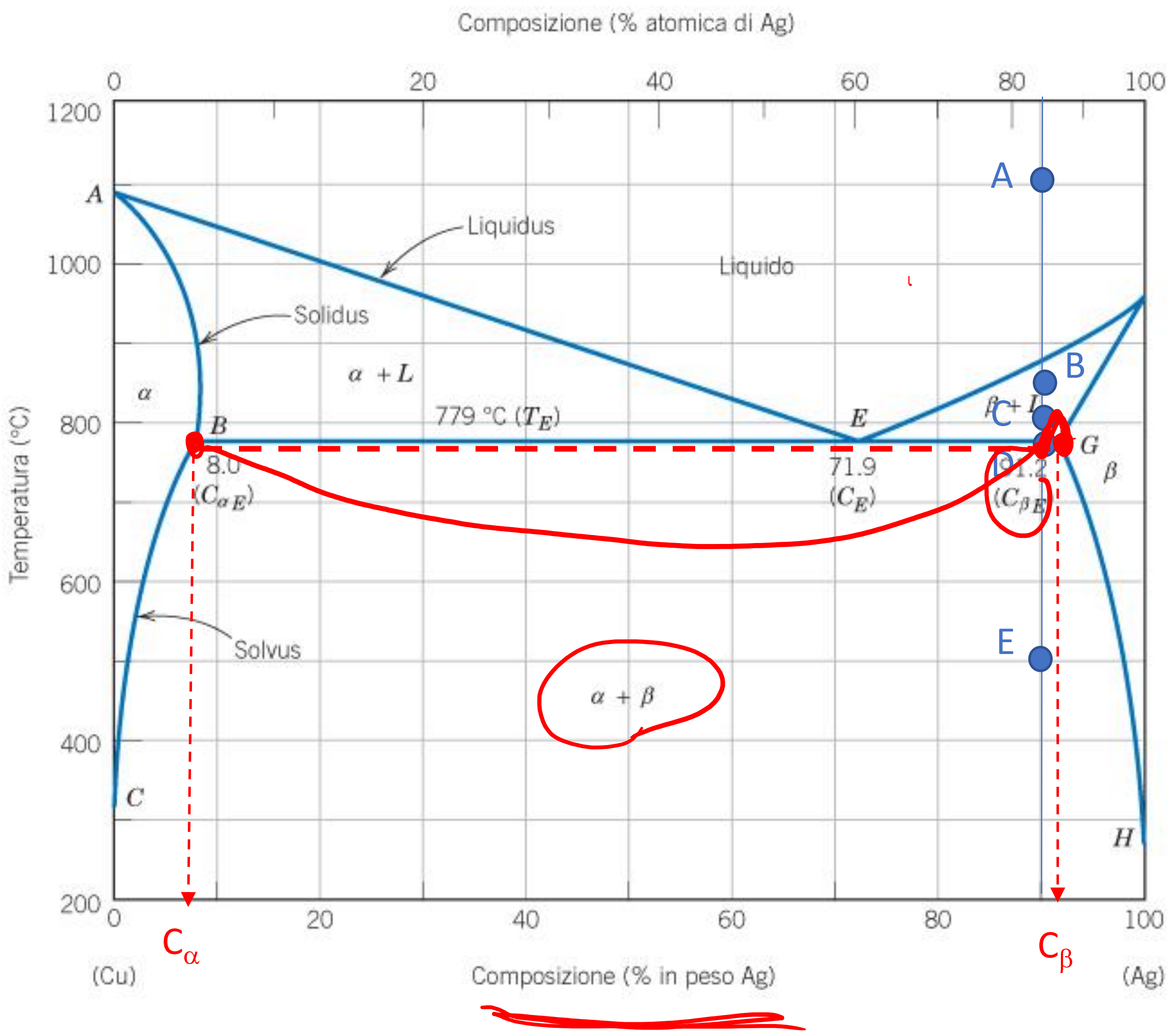
β	C	%
β	93% Ag 7% Cu	$\frac{90-85}{93-85}\cdot 100 =$ 62,5%
L	85% Ag 15% Cu	37,5%



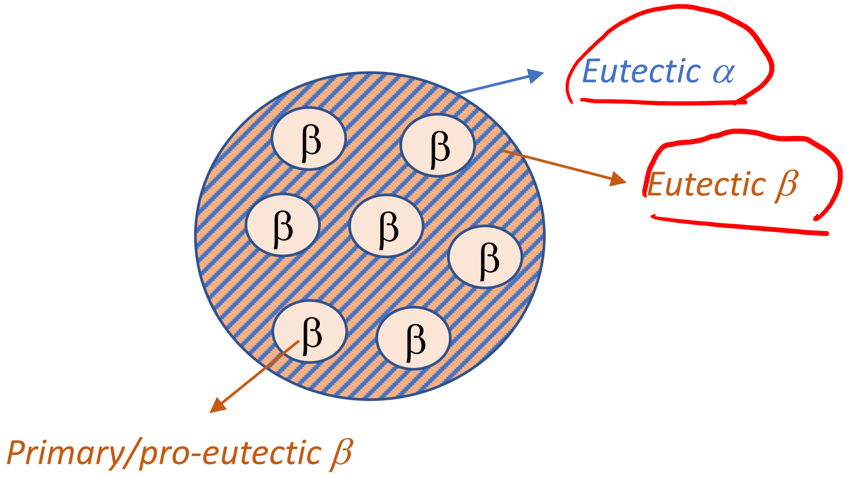


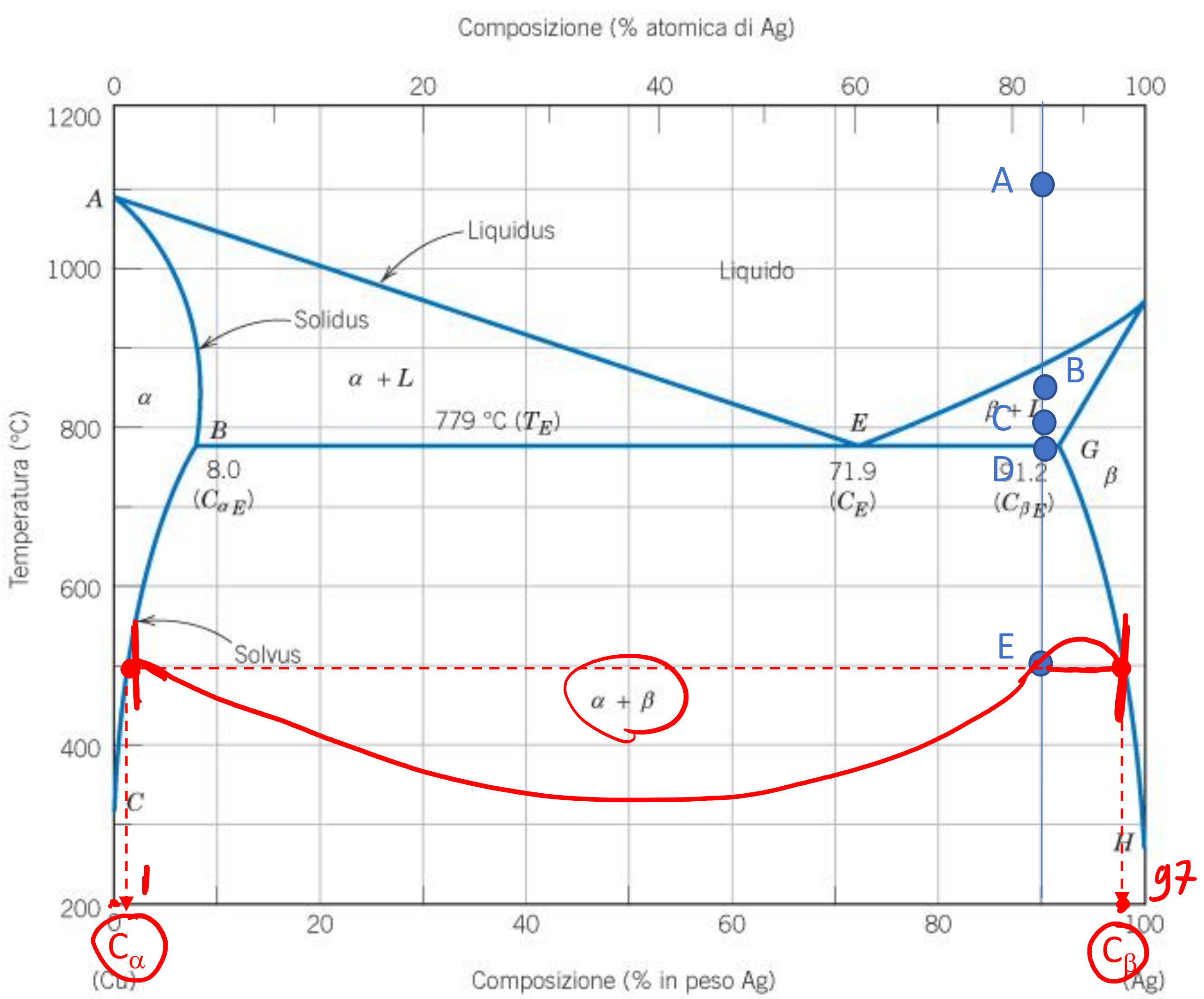
c	C	%
β	91,2 % Ag 8,8 % Cu	93,8 %
L	71,9 % Ag 28,1 % Cu	$\frac{91,2 - 90}{91,2 - 71,9} \cdot 100 =$ 6,2 %



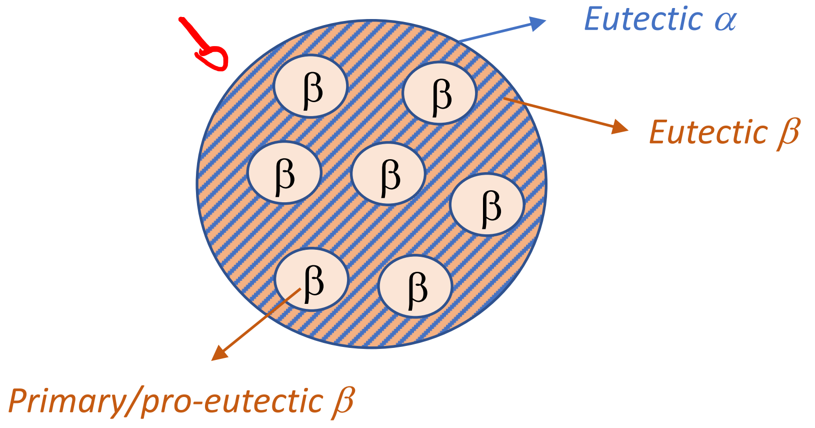


D	C	%
α	8% Ag 92% Cu	$\frac{91,2-90}{91,2-8} \cdot 100 =$ 1,4%
β	91,2% Ag 8,8% Cu	98,6%



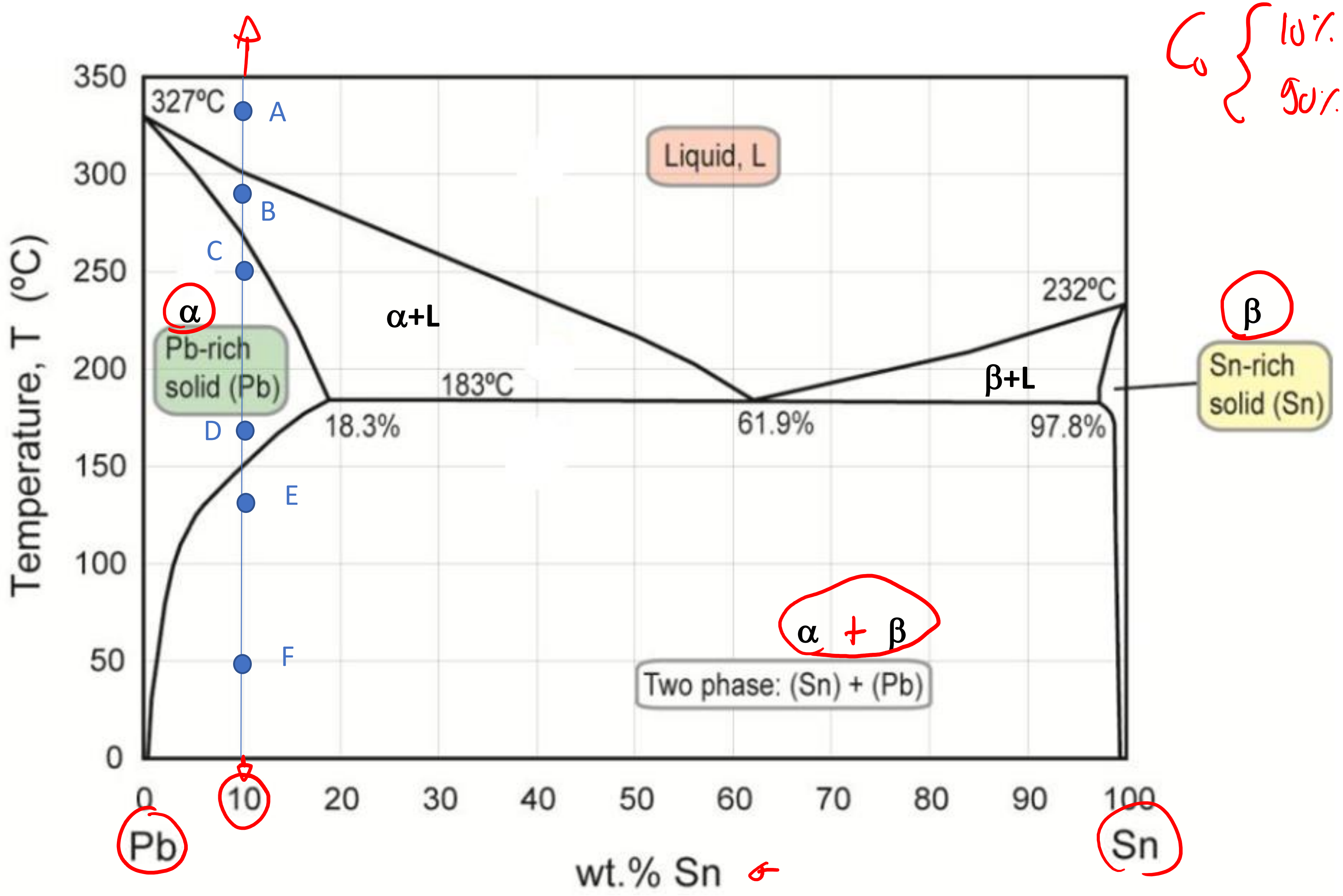


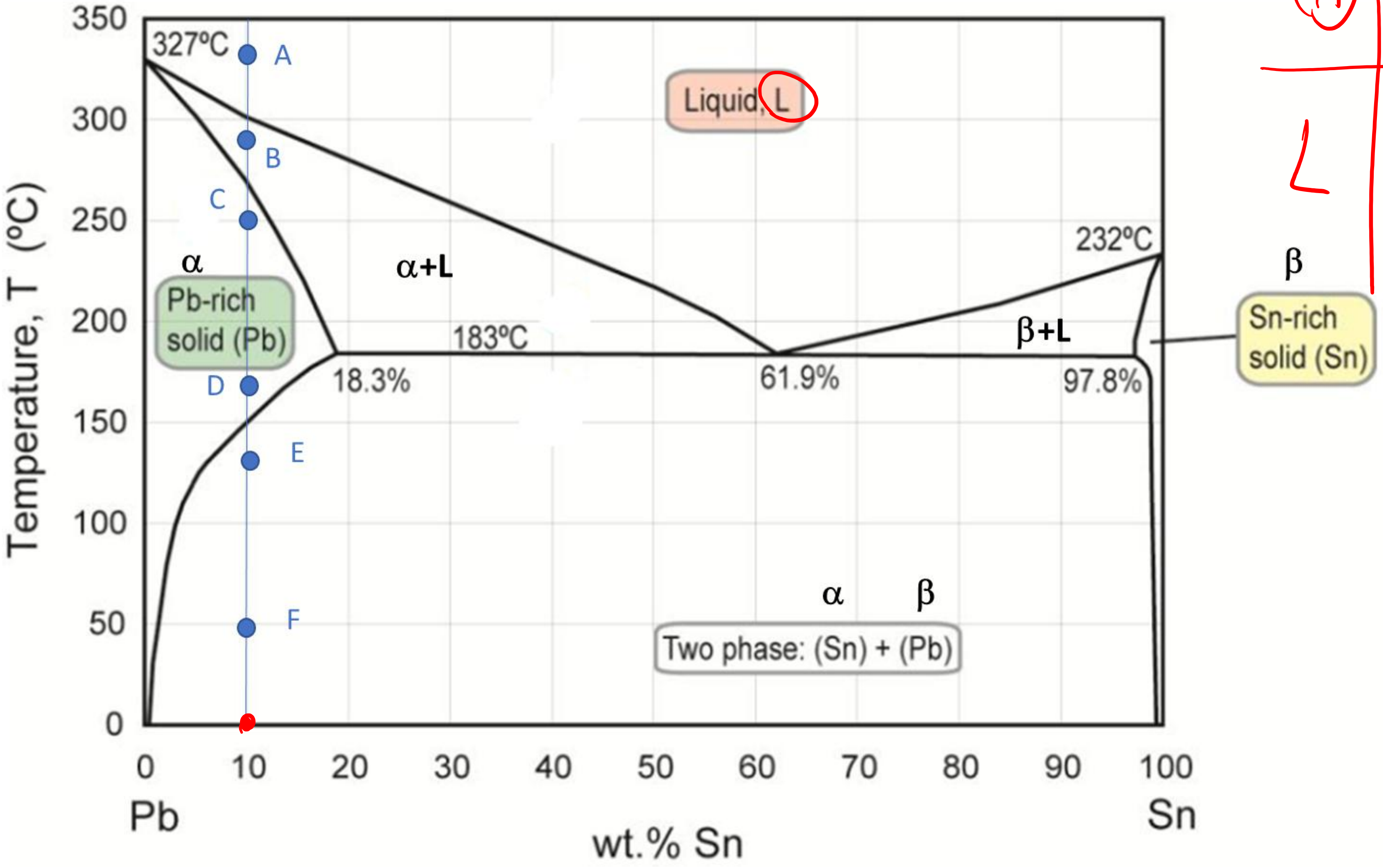
E	C	%
α	1% Ag 99% Cu	7,3%
β	97% Ag 3% Cu	$\frac{90-1}{97-1} \cdot 100 = 92{,}7\%$



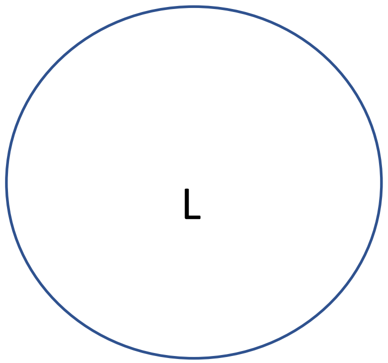
Exercise 2: determine the phase present, their composition and relative amounts in the points A, B, C, D, E and F indicated in the phase diagram. Schematize the microstructure in each point.

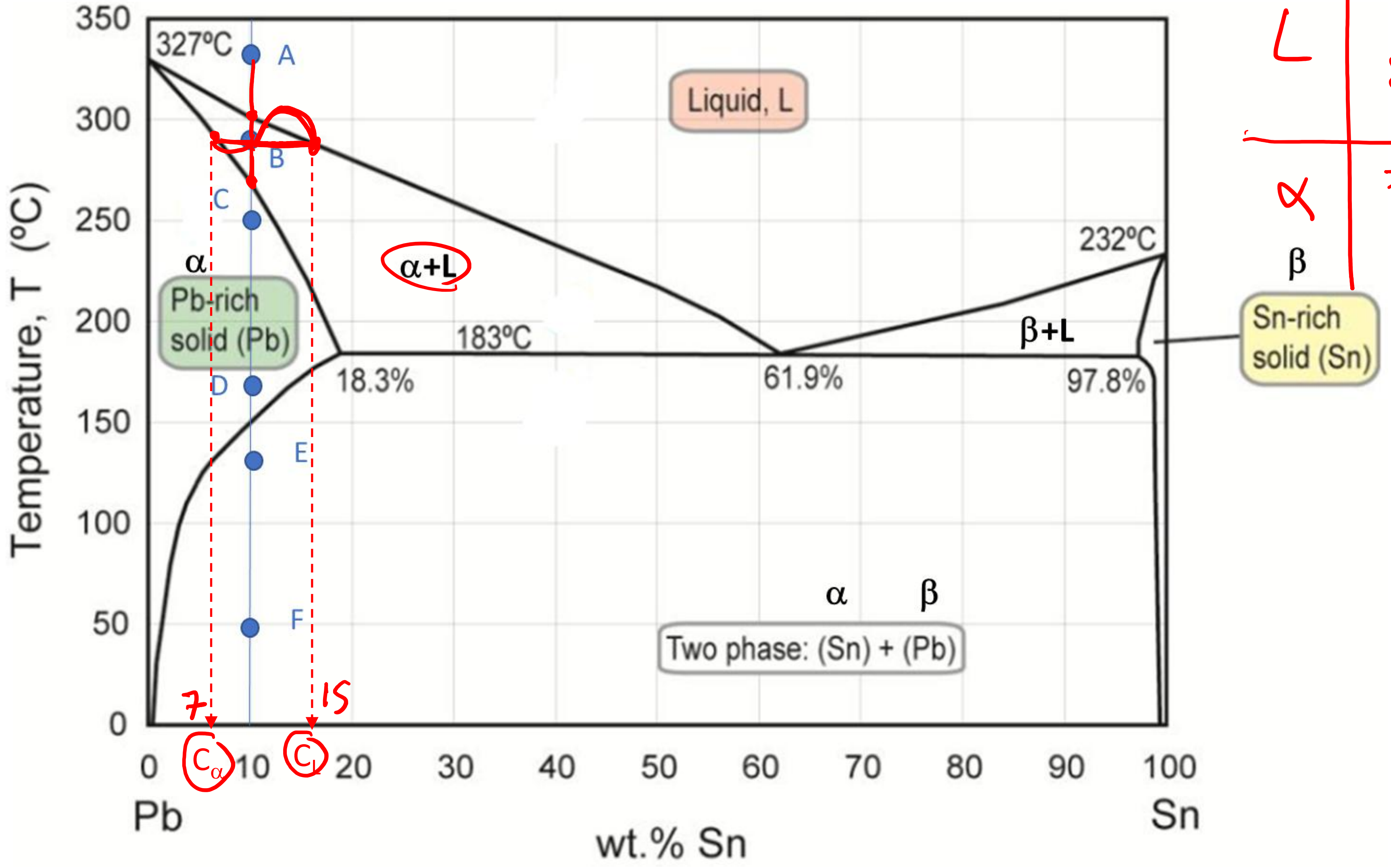
$C_0 \begin{cases} 10\%\ Sn \\ 90\%\ Pb \end{cases}$



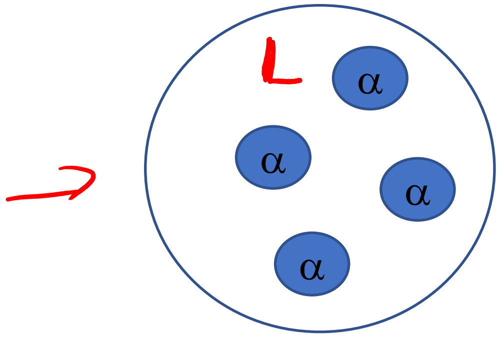


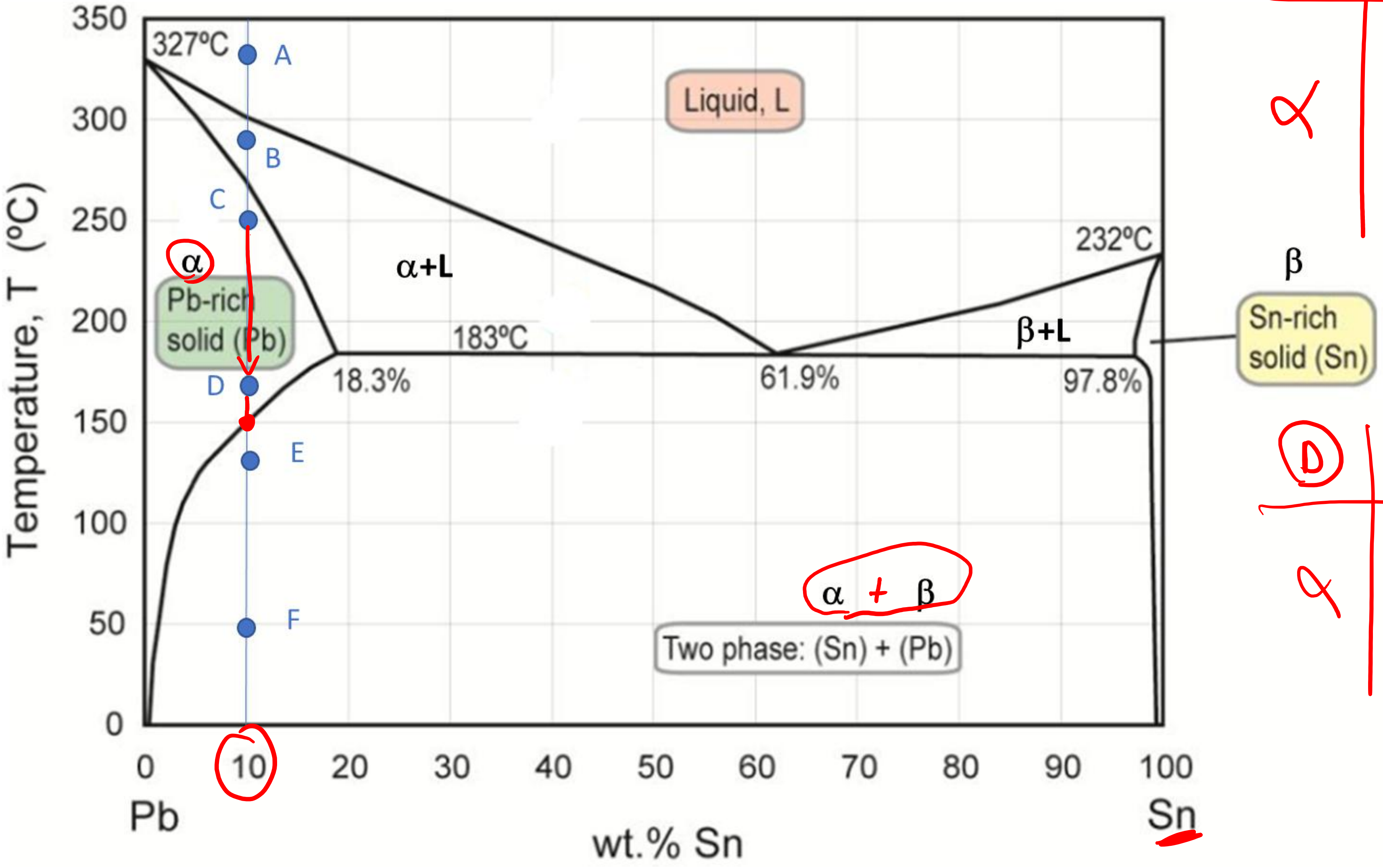
A	C	%
L	10% Sn 90% Ag	100%





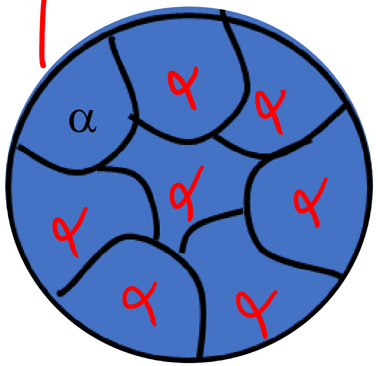
B	C	%
L	15% Sn 85% Pb	37,5%
α	7% Sn 93% Pb	$\frac{15-10}{15-7} \cdot 100 =$ 62,5%

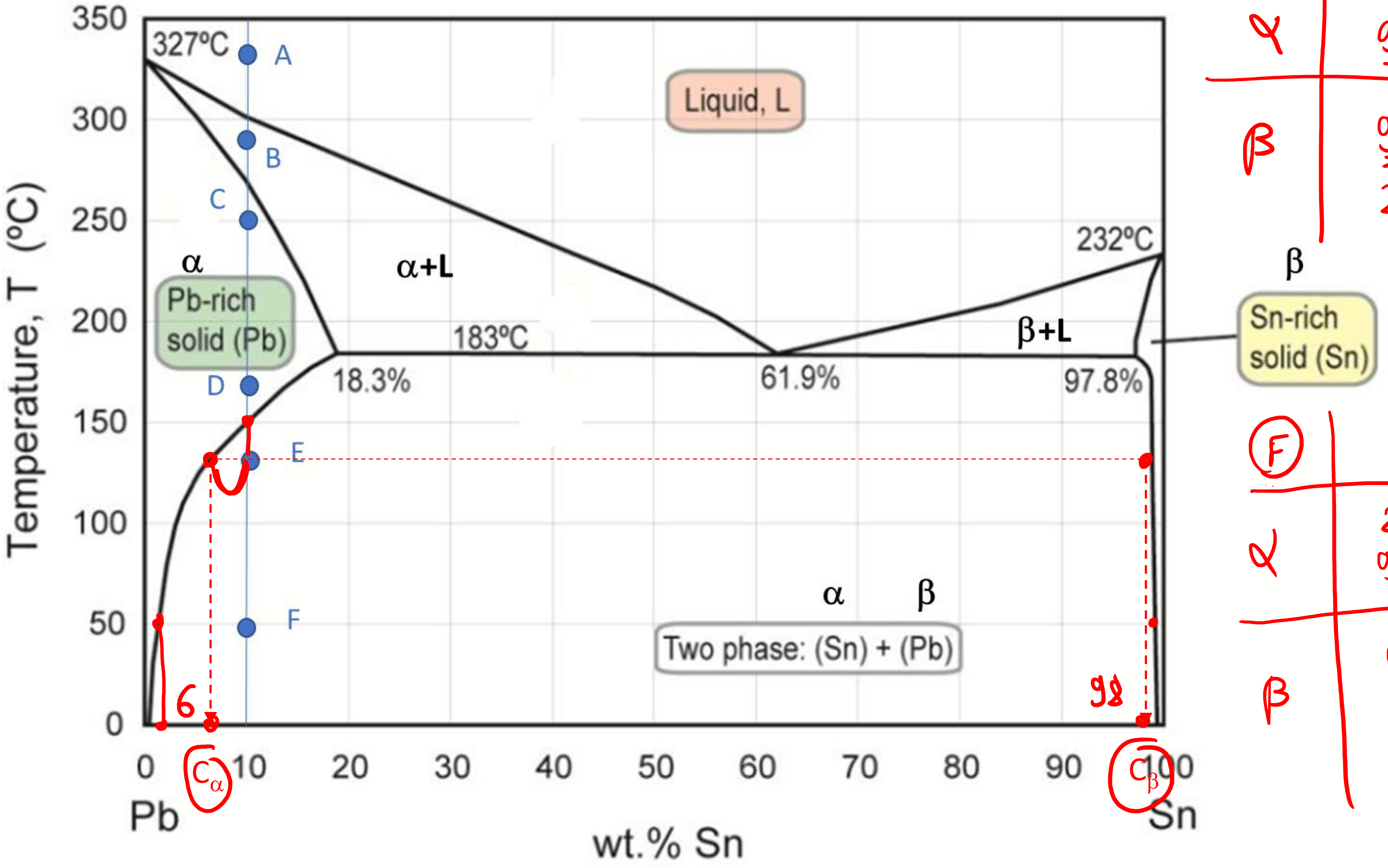




C	C	%
α	10% Sn 90% Pb	100%

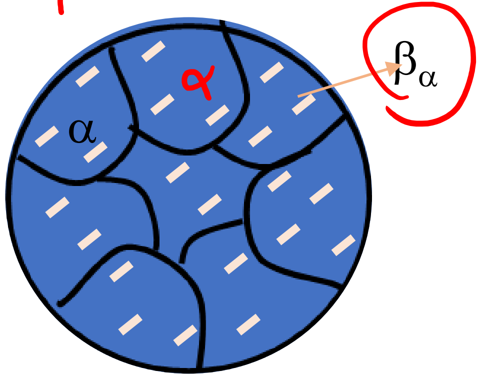
D	C	%
α	10% Sn 90% Pb	100%

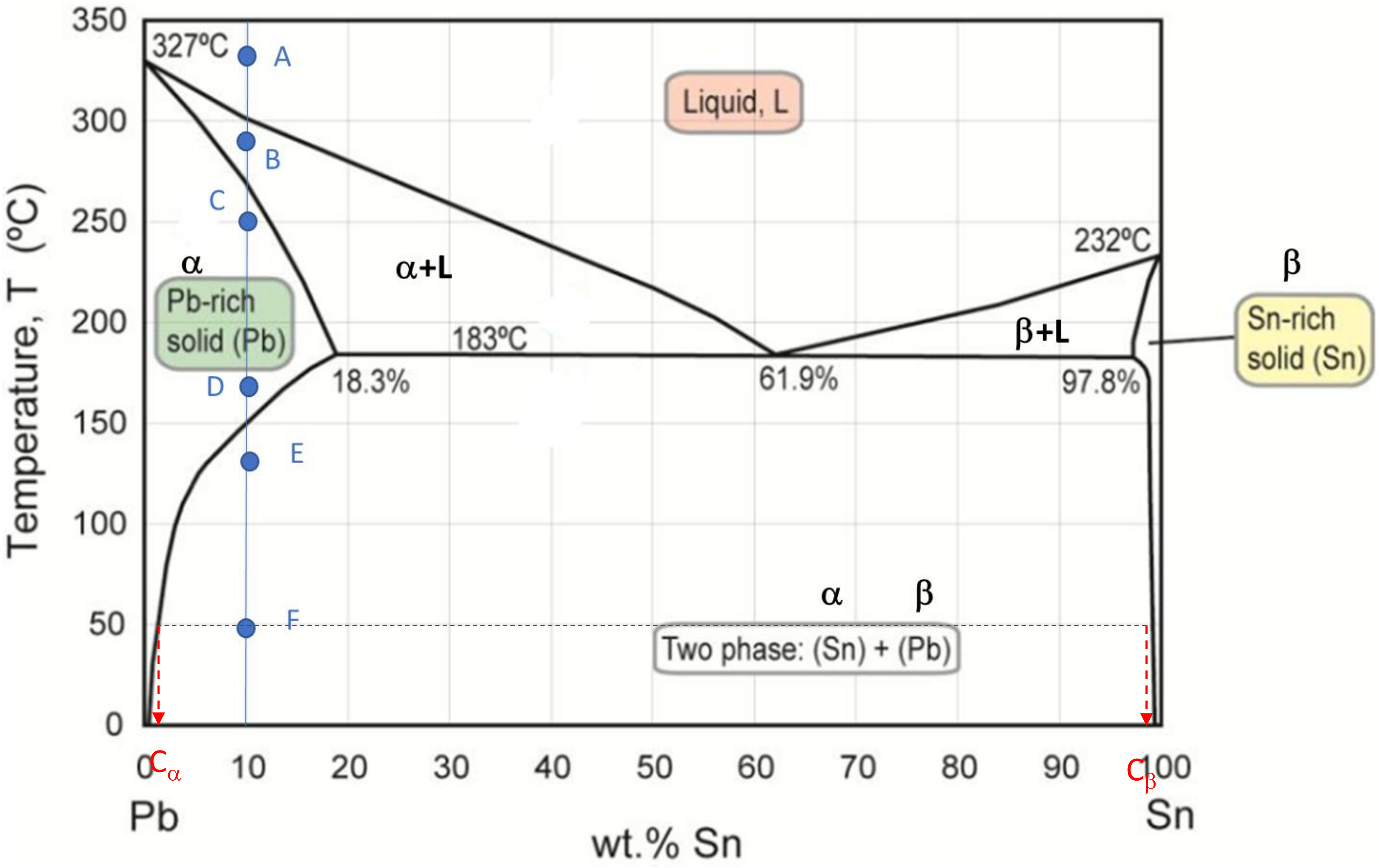




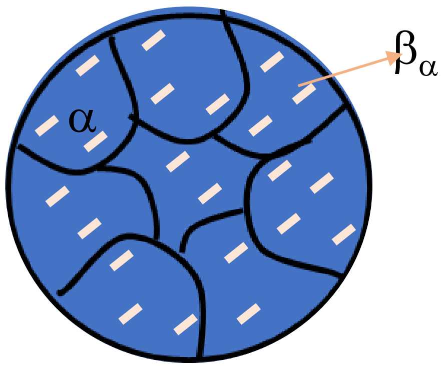
E	C	%
α	6% Sn 94% Pb	96%
β	98% Sn 2% Pb	$\frac{10-6}{98-6} \cdot 100 = 4\%$

F	C	%
α	2% Sn 98% Pb	92%
β	98% Sn 2% Pb	8%



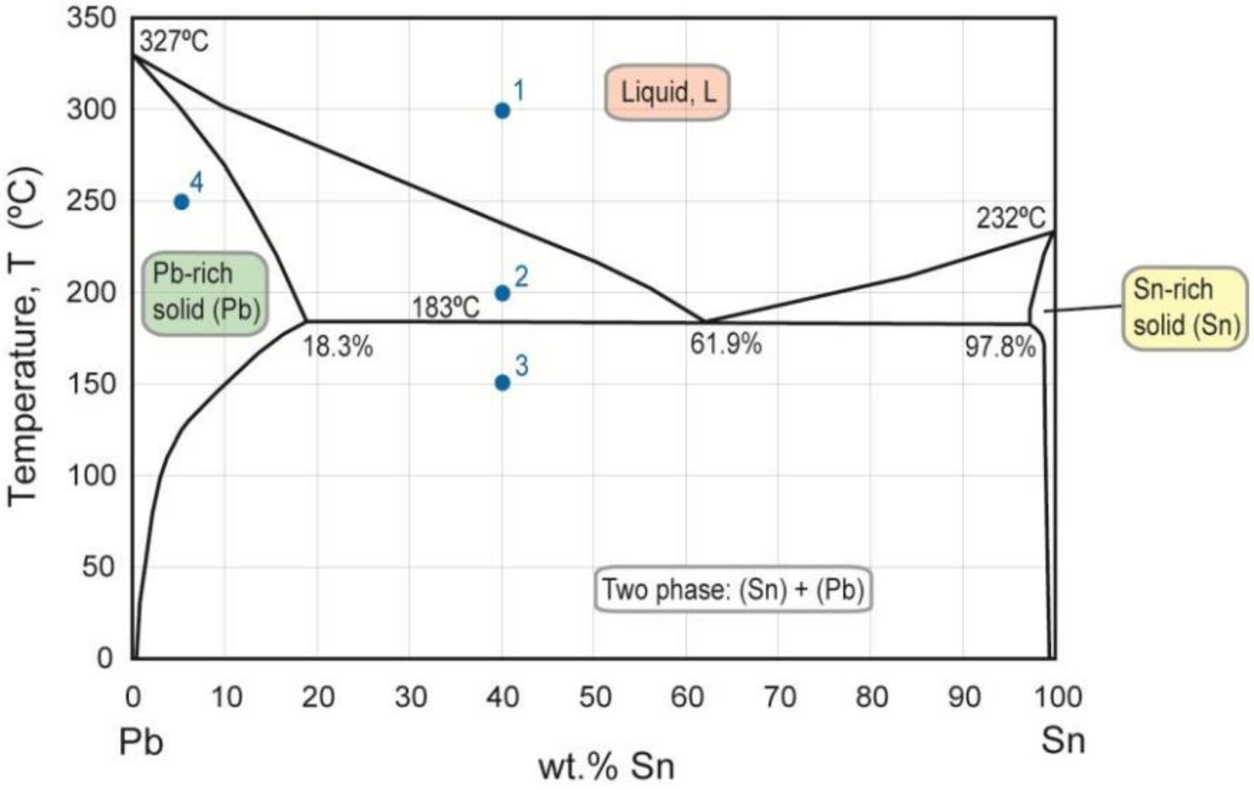


350
300
250
200
150
100
50
0
Temperature, T (ºC)
327ºC
A
B
C
D
E
F
Liquid, L
α
Pb-rich
solid (Pb)
α+L
183ºC
18.3%
61.9%
97.8%
232ºC
β
β+L
Sn-rich
solid (Sn)
α
β
Two phase: (Sn) + (Pb)
0
10
20
30
40
50
60
70
80
90
100
Pb
Sn
wt.% Sn
C_α
C_β



β_α
α

E.6 Use the Pb-Sn diagram below to answer the following questions.



(a) What are the values of the state variables (composition and temperature) at constitution point 1?

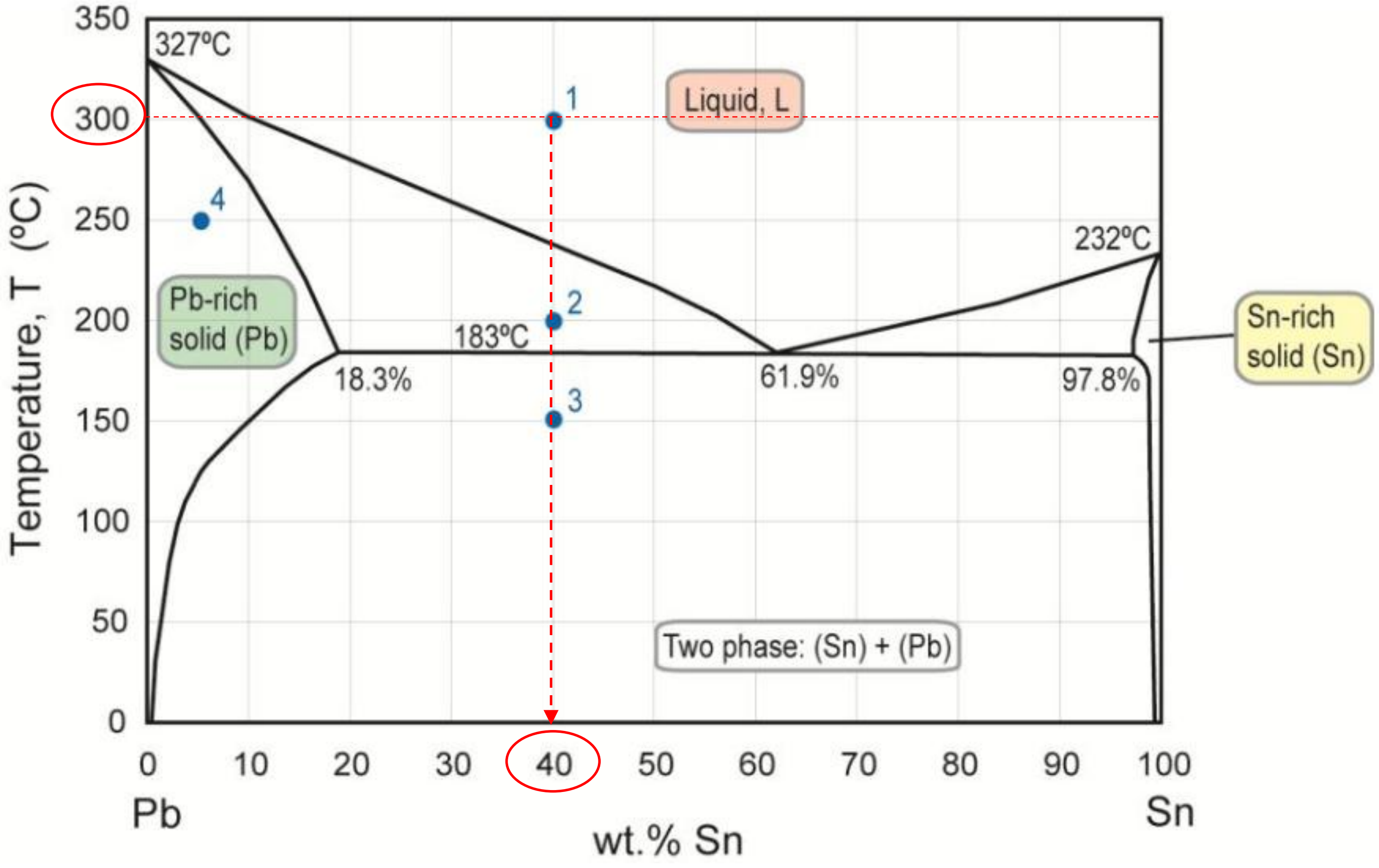
(b) Mark the constitution points for Pb-60wt% Sn and Pb-20wt% Sn alloys at 250°C. What phases are present in each case?

(c) The alloy at constitution point 1 is cooled very slowly to room temperature, maintaining equilibrium. At which temperatures do changes in the number or type of phases occur? What phases are present at constitution points 2 and 3?

(d) The alloy at constitution point 4 is cooled slowly to room temperature. Identify the following:

(i) the initial composition, temperature and phase(s);

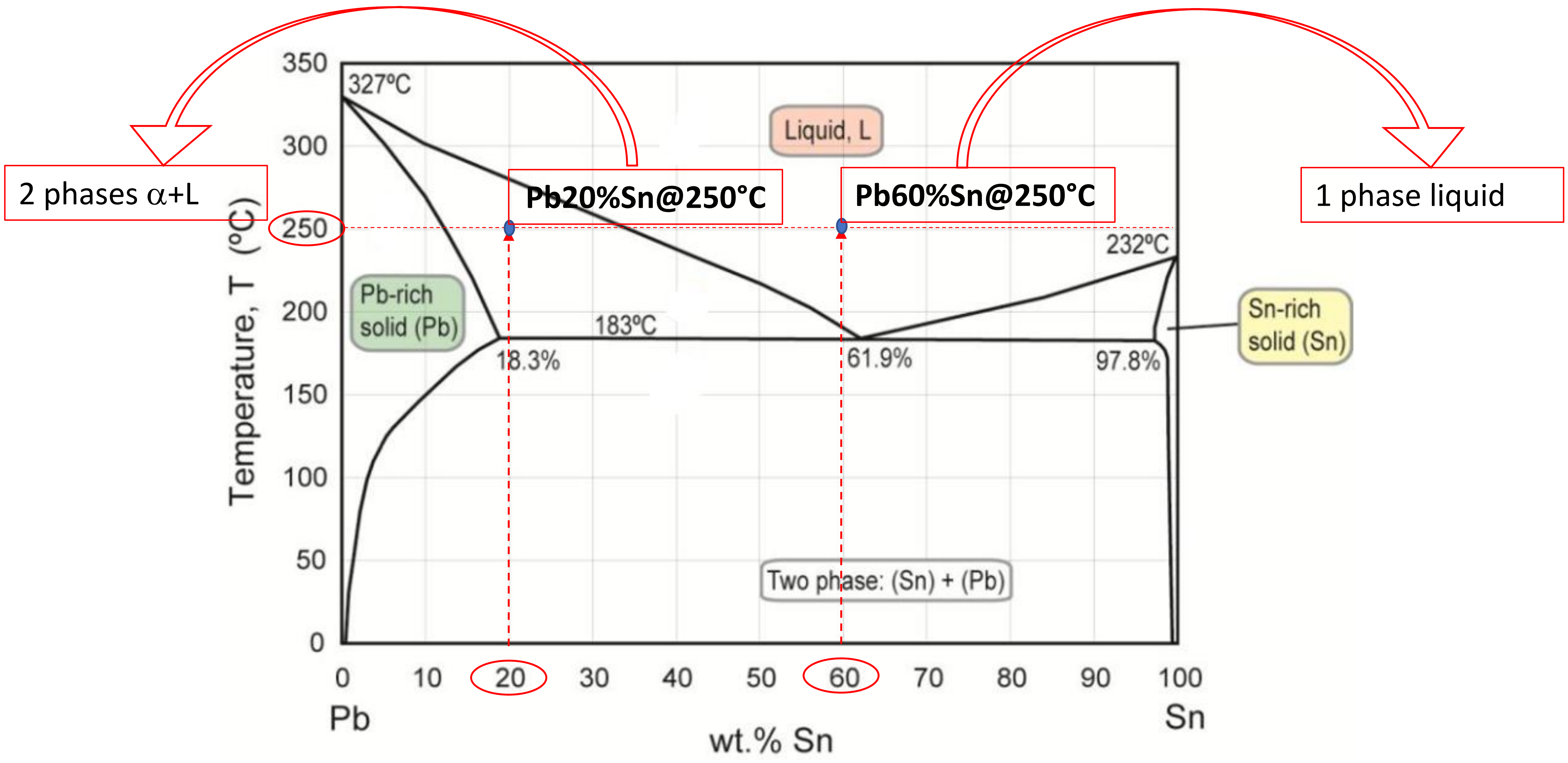
(ii) the temperature at which a phase change occurs, and the final phase(s).



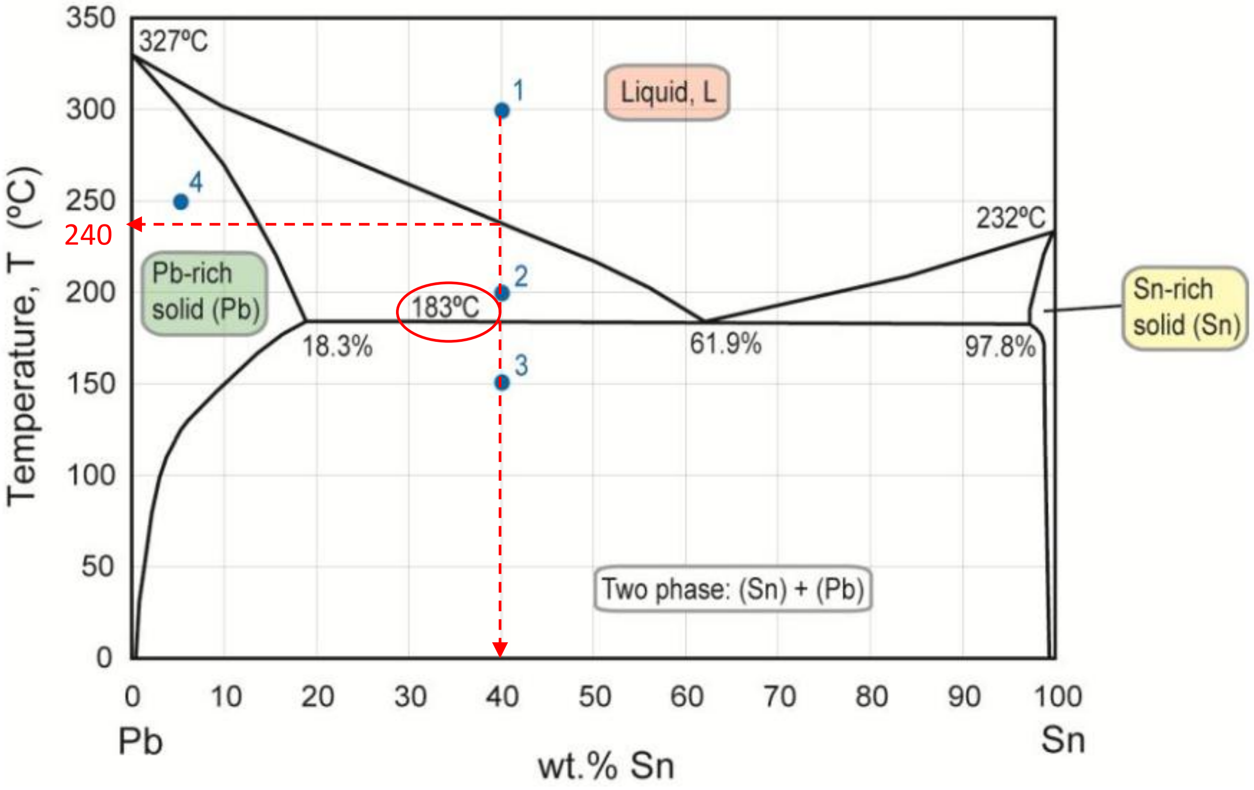
Point 1

1 phase liquid, composition 40%Sn, 60% Pb

Temperature 300°C


2 phases α+L
Pb20%Sn@250°C
Pb60%Sn@250°C
1 phase liquid
327ºC
Liquid, L
232ºC
Pb-rich
solid (Pb)
183ºC
18.3%
61.9%
97.8%
Sn-rich
solid (Sn)
Two phase: (Sn) + (Pb)
Temperature, T (ºC)
350
300
250
200
150
100
50
0
0
10
20
30
40
50
60
70
80
90
100
Pb
Sn
wt.% Sn

(c) The alloy at constitution point 1 is cooled very slowly to room temperature, maintaining equilibrium. At which temperatures do changes in the number or type of phases occur? What phases are present at constitution points 2 and 3?



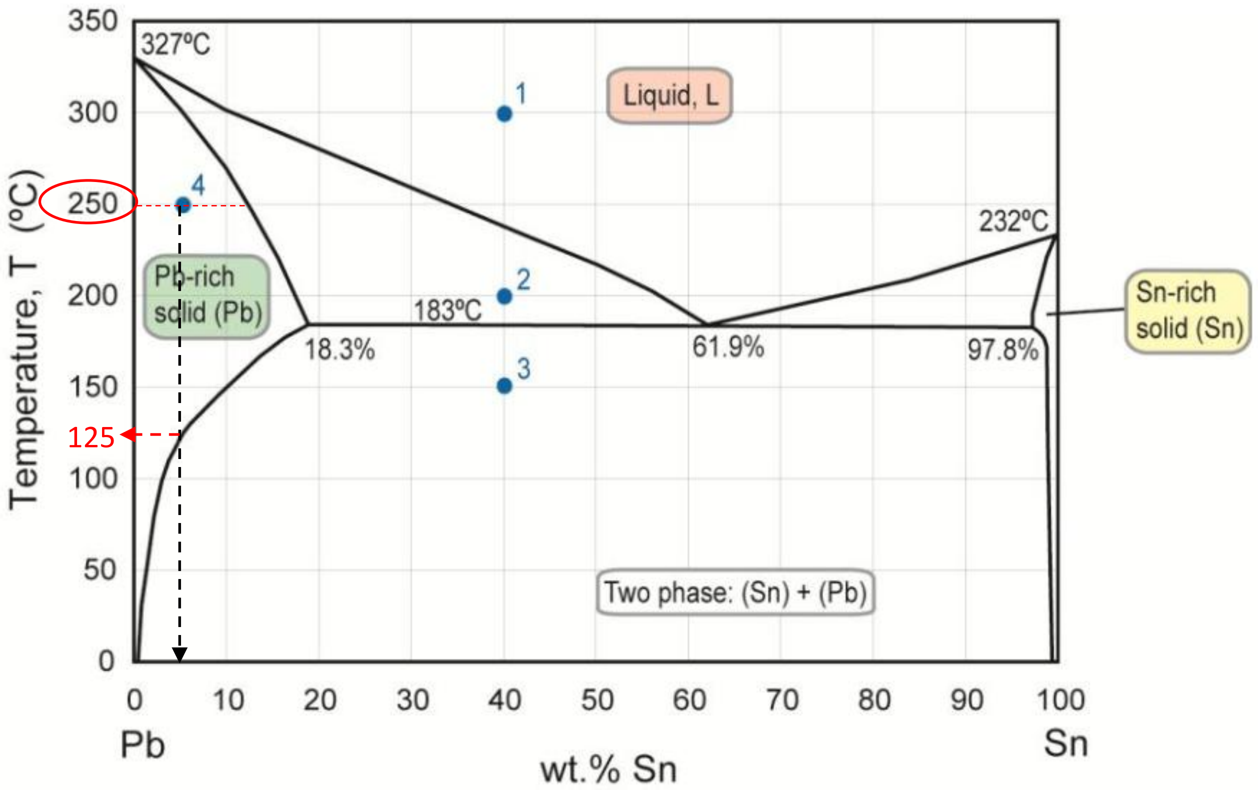
Alloy 1:

- 300°C – only liquid
- 240°C - solidification of the first α crystal
- 183°C - all the liquid undergoes eutectic transformation producing α-β lamellar microstructure

Point 2: 2 phases α (solid rich in Pb) and Liquid (L)

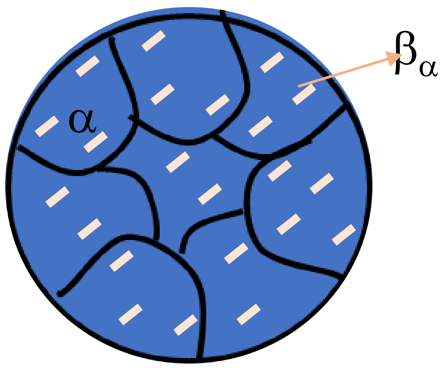
Point 3: 2 phases α (solid rich in Pb) and β (solid rich in Sn)

(d) The alloy at constitution point 4 is cooled slowly to room temperature. Identify the following:
(i) the initial composition, temperature and phase(s);
(ii) the temperature at which a phase change occurs, and the final phase(s).

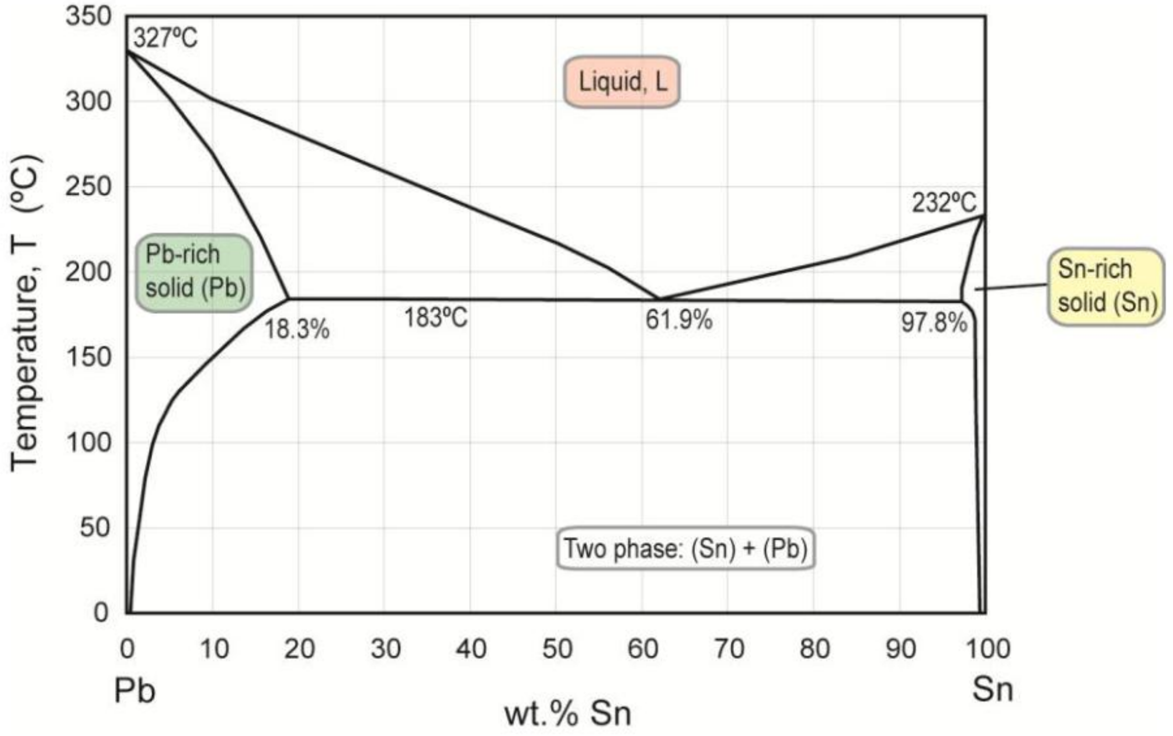


Alloy 4:
Start – 250°C, α (solid phase rich in Pb), composition : 5%Sn, 95%Pb

125°C precipitation of β_α solid phase (rich in Sn) due to reduction of the solubility of Sn in Pb at that temperature
From 125°C to room temperature no other phase changes occur, the microstructure is solid phase α (rich in Pb) with small β_α crystals



E.7 Use the Pb-Sn diagram below to answer the following questions.

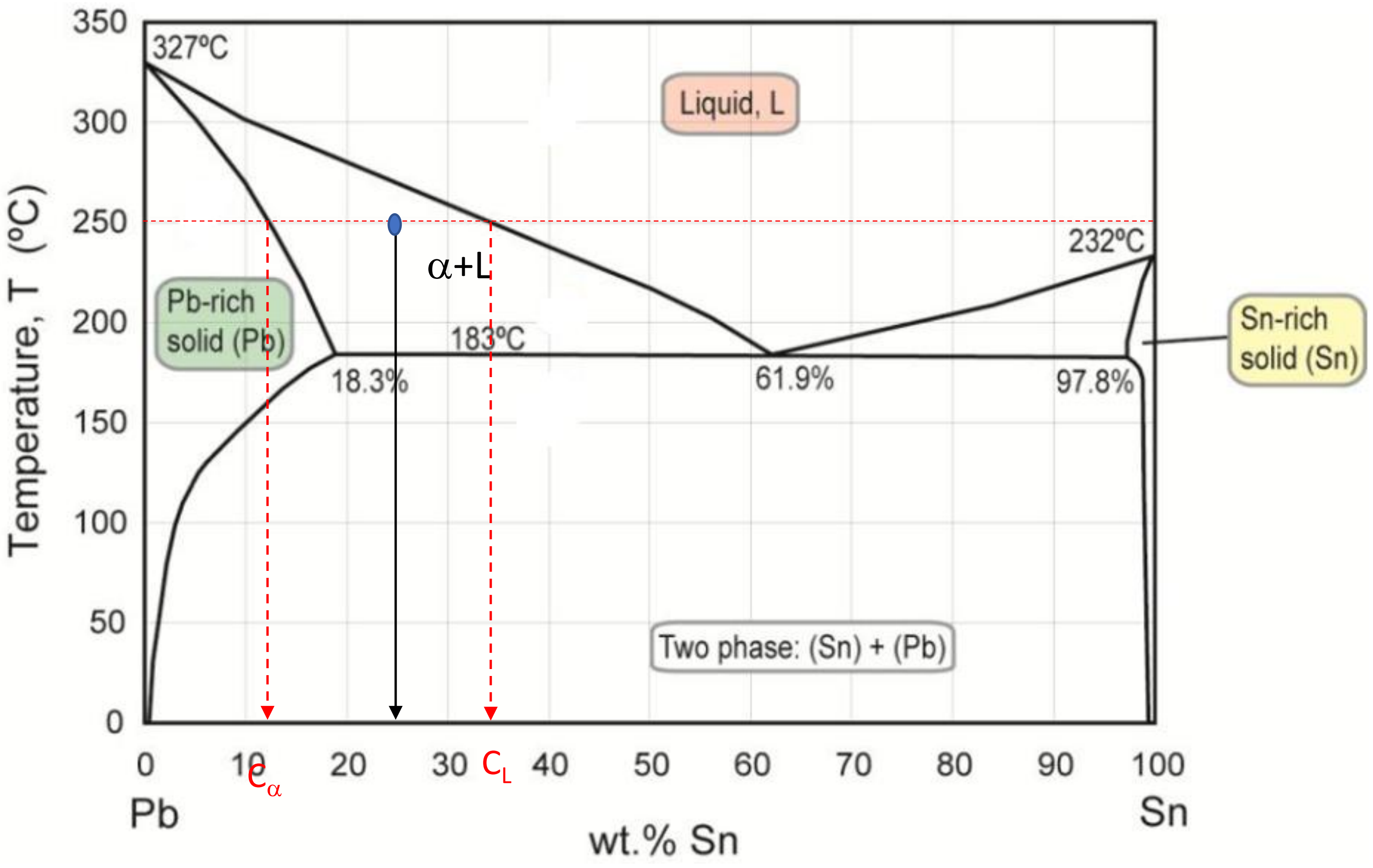


(a) The constitution point for a Pb-25wt% Sn alloy at 250 °C lies in a two-phase field. Construct a tie-line on the figure and read off the two phases and their compositions.

(b) The alloy is slowly cooled to 200 °C. At this temperature, identify the phases and their compositions

(c) The alloy is cooled further to 150 °C. At this temperature, identify the phases and their compositions.

(d) Indicate with arrows on the figure the lines along which the compositions of phase 1 and phase 2 move during slow cooling from 250°C to 200°C.

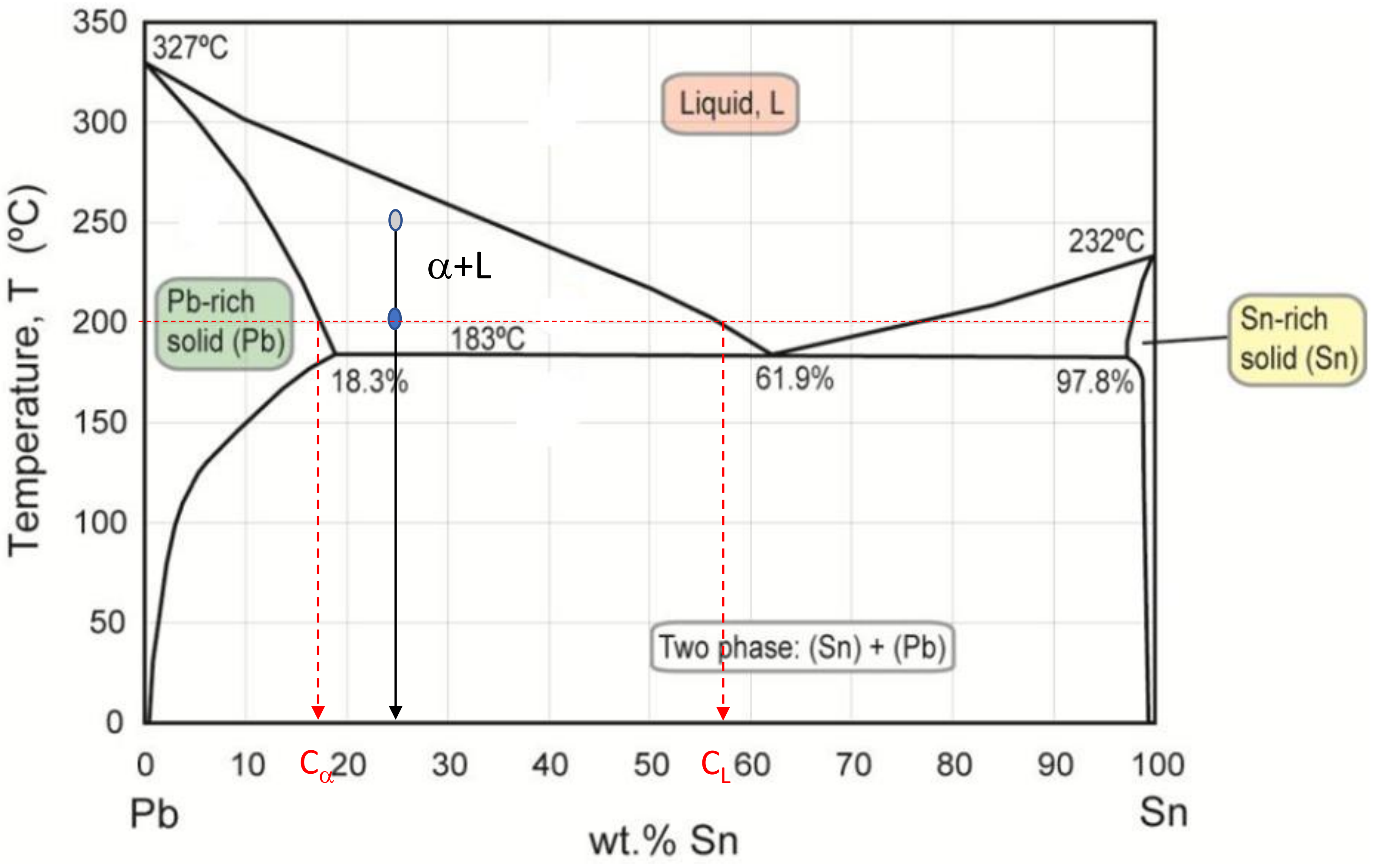


Alloy 25%Sn 75%Pb @ 250°C

α (solid phase rich in Pb) and liquid (L)

Composition of α=12%Sn, 88%Pb

Composition of liquid: 33%Sn, 67%Pb

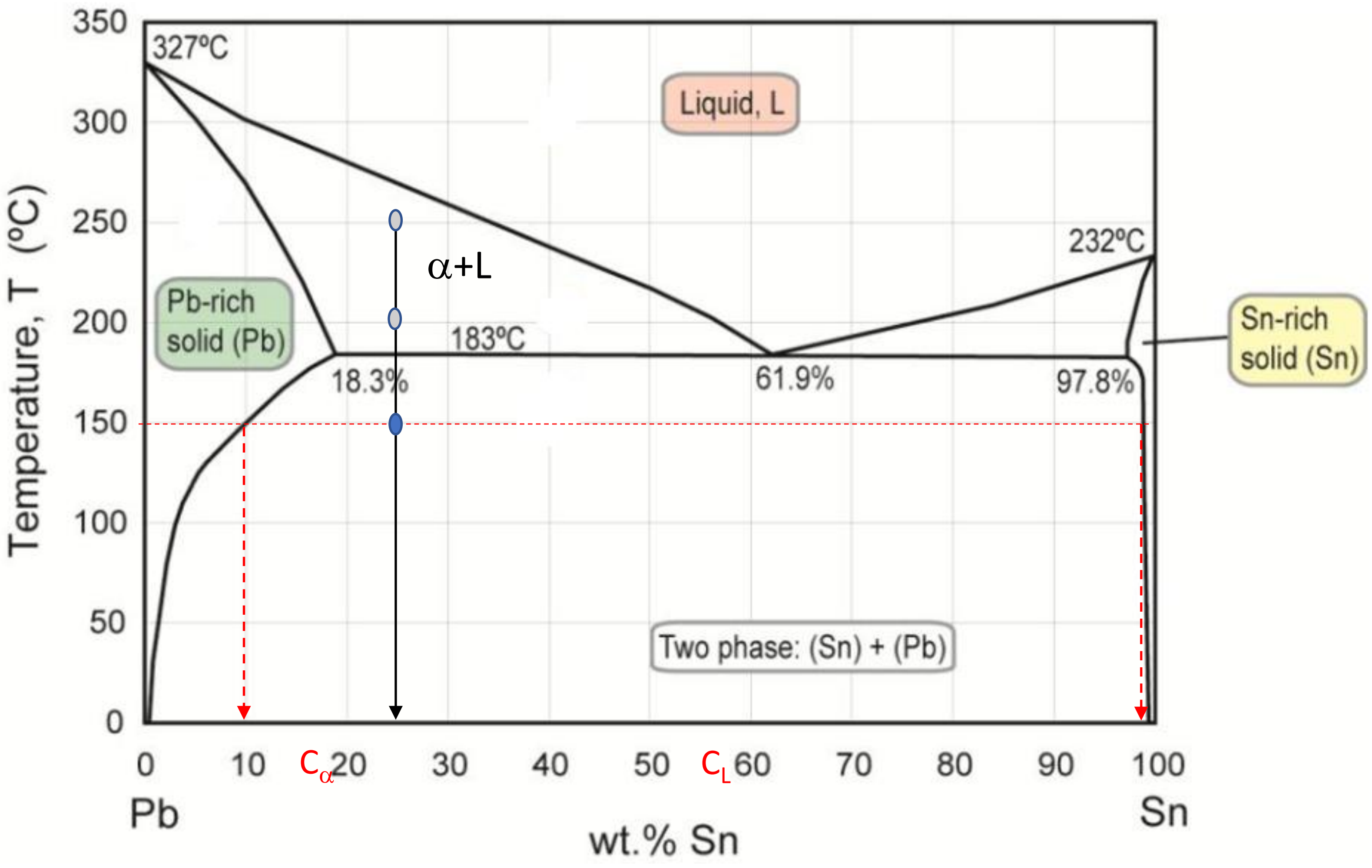


Alloy 25%Sn 75%Pb @ 200°C

α (solid phase rich in Pb) and liquid (L)

Composition of α=18%Sn, 82%Pb

Composition of liquid: 58%Sn, 42%Pb

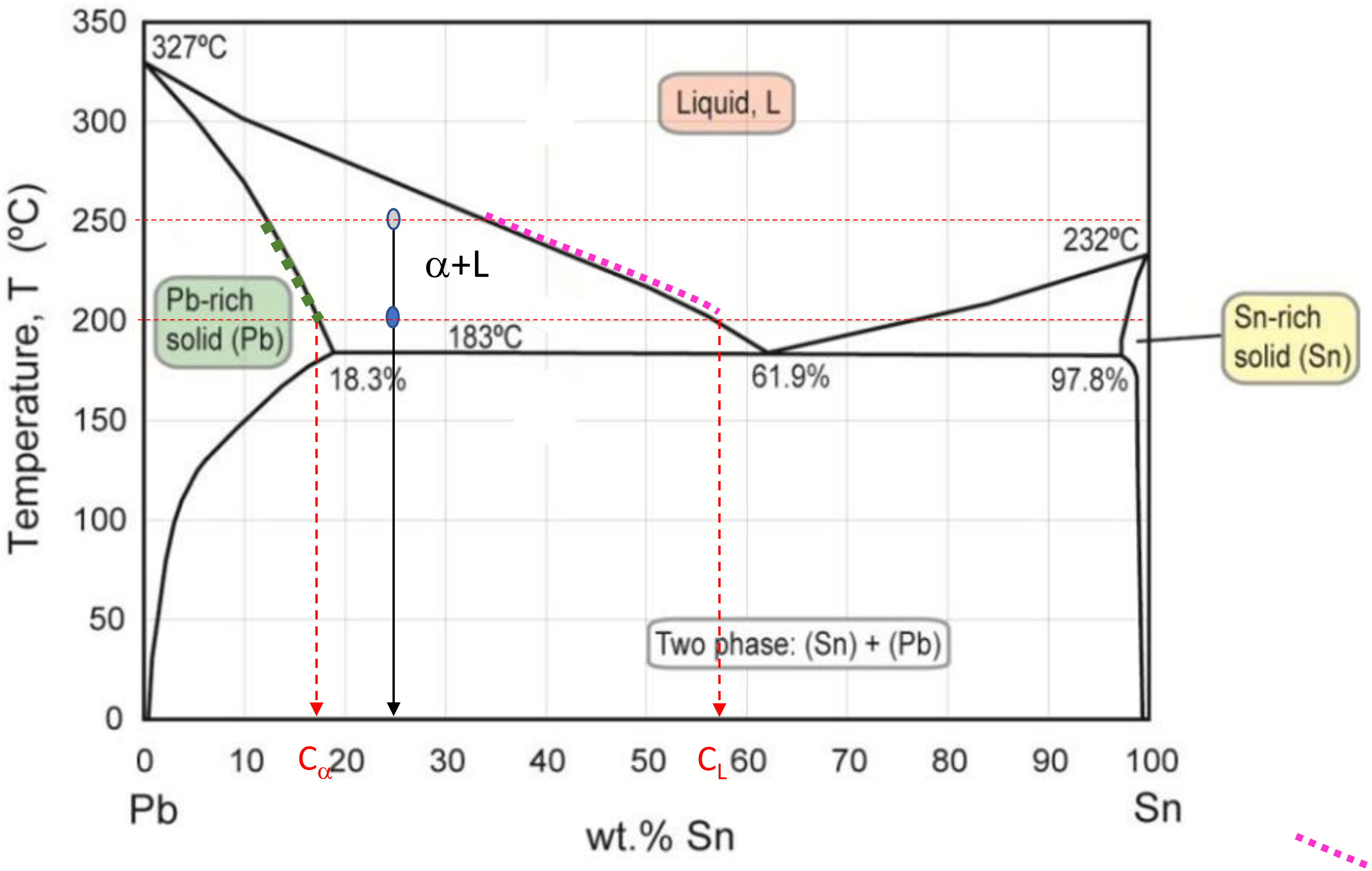


Alloy 25%Sn 75%Pb @ 150°C

α (solid phase rich in Pb) and β (solid rich in Sn)

Composition of α=10%Sn, 90%Pb

Composition of β: 99%Sn, 1%Pb



Alloy 25%Sn 75%Pb @ 250°C

α (solid phase rich in Pb) and liquid (L)

Composition of α=12%Sn, 88%Pb

Composition of liquid: 33%Sn, 67%Pb

Alloy 25%Sn 75%Pb @ 200°C

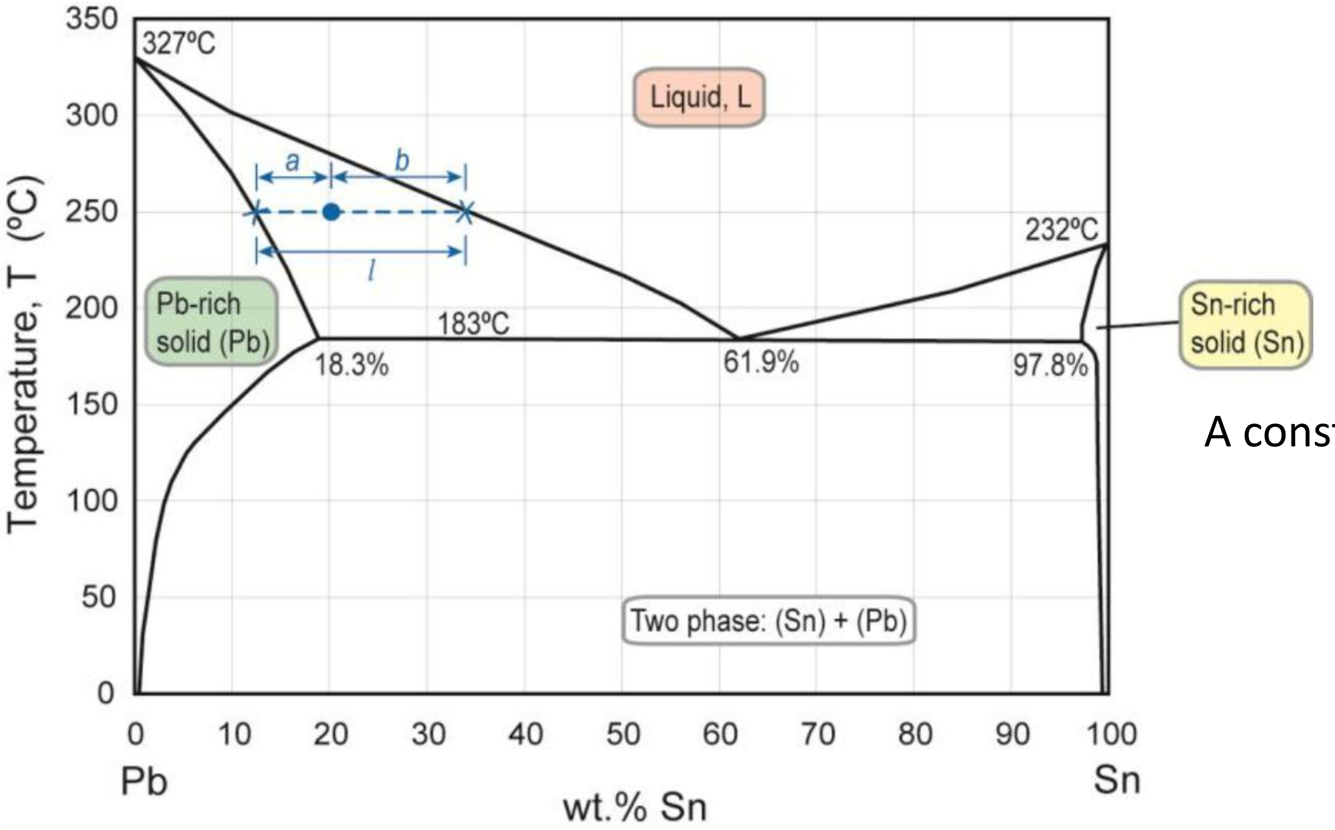
α (solid phase rich in Pb) and liquid (L)

Composition of α=18%Sn, 82%Pb

Composition of liquid: 58%Sn, 42%Pb

Variation of α composition reducing temperature from 250 to 200 °C

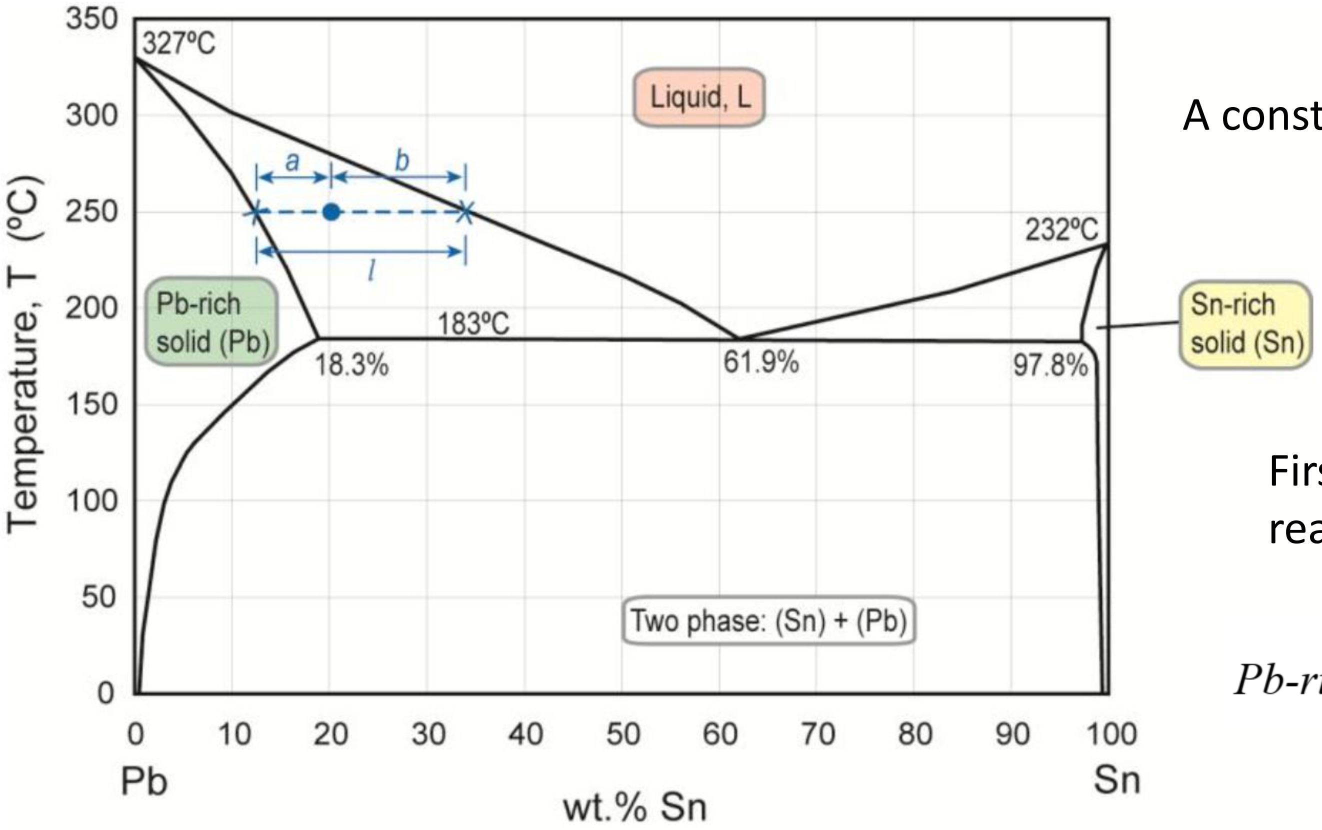
Variation of L composition reducing temperature from 250 to 200 °C



Consider Pb-20wt% Sn at 250 ^{o}C in Figure

A constitution point in the two-phase field: liquid plus Pb-rich solid.

Find the proportions of each phase



Consider Pb-20wt% Sn at 250 ^oC in Figure

A constitution point in the two-phase field: liquid plus Pb-rich solid.

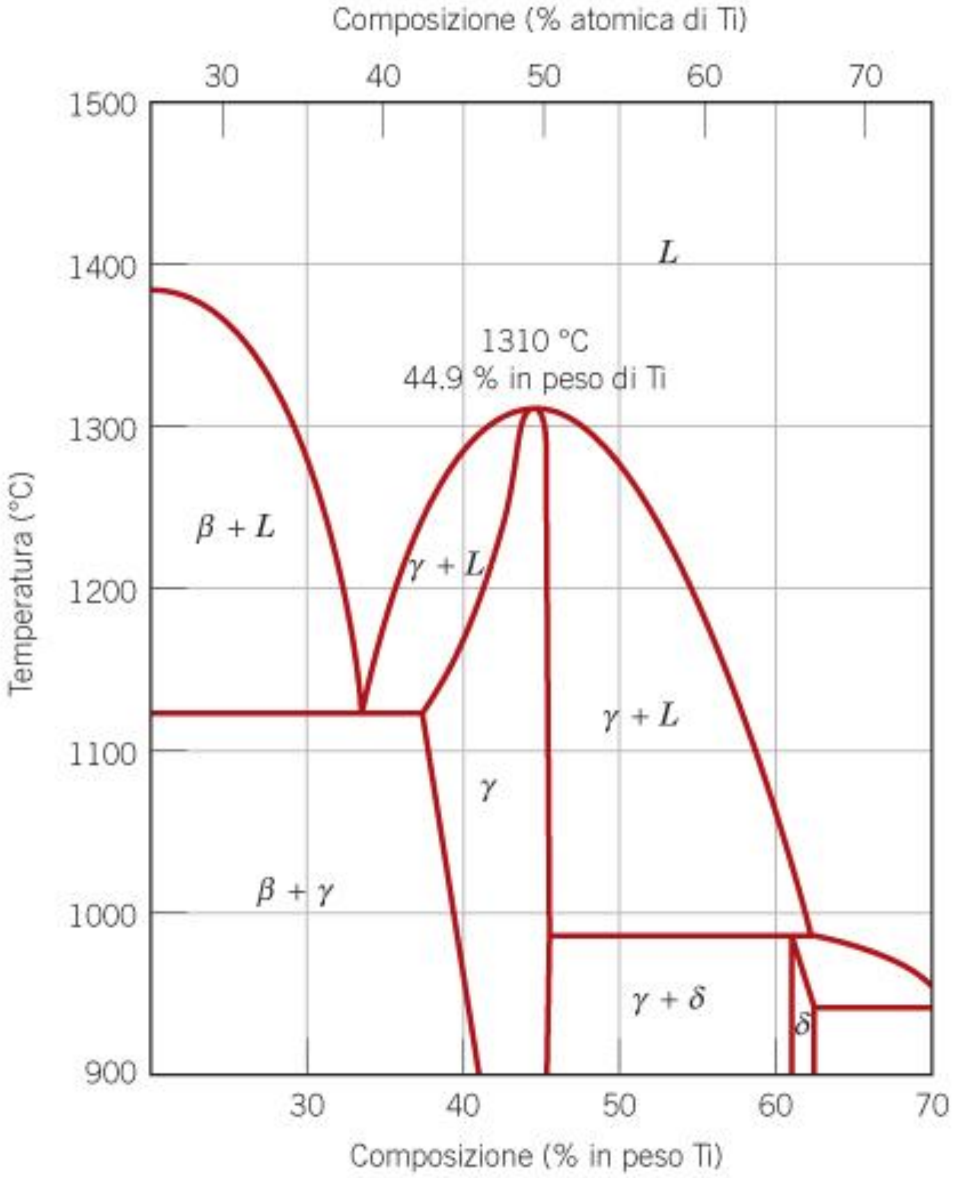
To find the proportions of each phase

First construct a tie-line through the constitution point and read off the compositions of the phases:

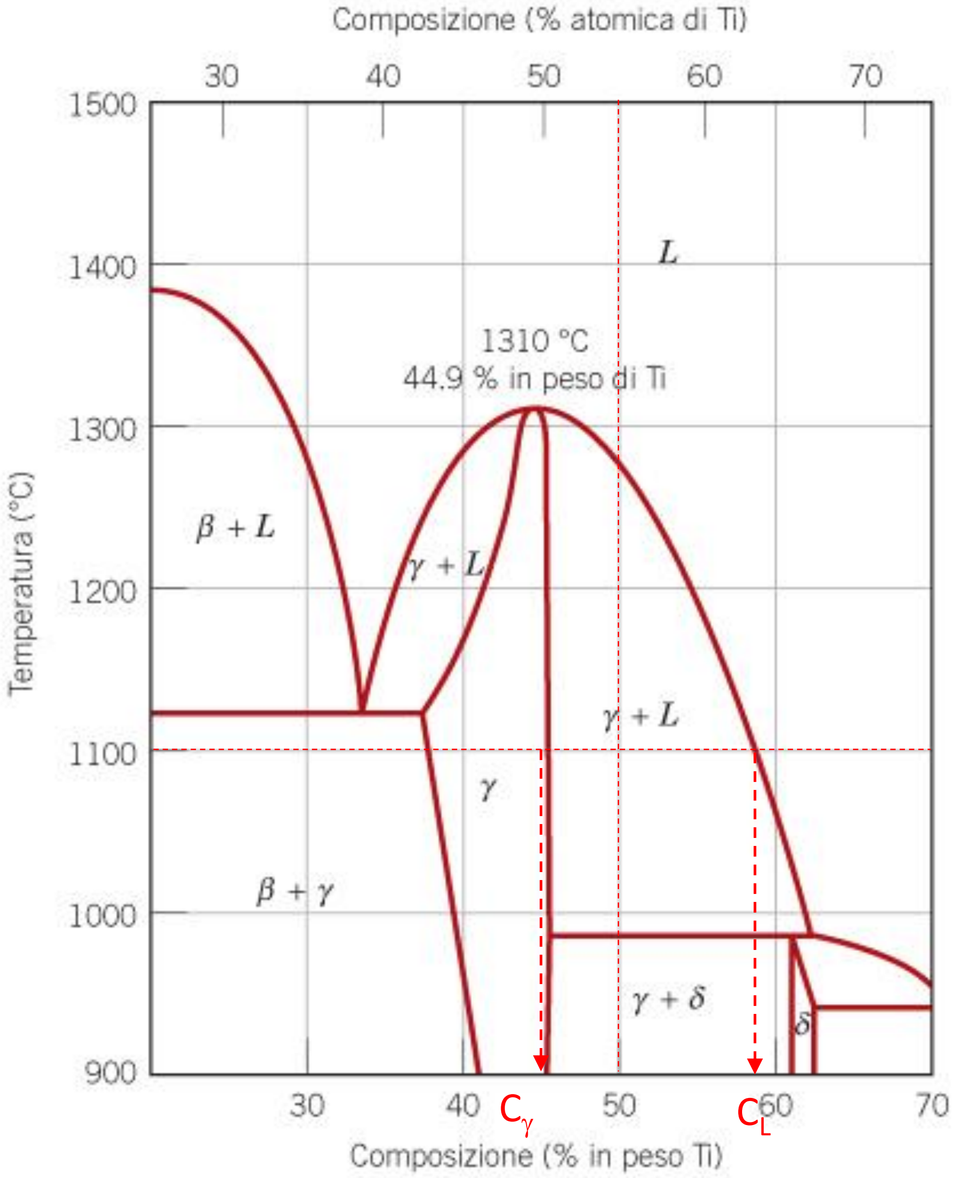
$$\textit{Pb-rich solid} \text{ with } W_{Sn}^{SOL} = 12\%; \quad \textit{Liquid} \text{ with } W_{Sn}^{LIQ} = 34\%.$$

The tie-line is of length l , while the lengths of the segments to either side of the constitution point are **a** and **b** respectively

For the example alloy of composition W Sn = 20%



Determine the phase/s present, the composition/s and relative amount/s for an alloy with 50%Ti at 1100°C.



Determine the phase/s present, the composition/s and relative amount/s for an alloy with 50%Ti at 1100°C.

Phases: solid phase γ (intermediate solid solution) and liquid L

Compositions:

Composition of γ: 45%Ti

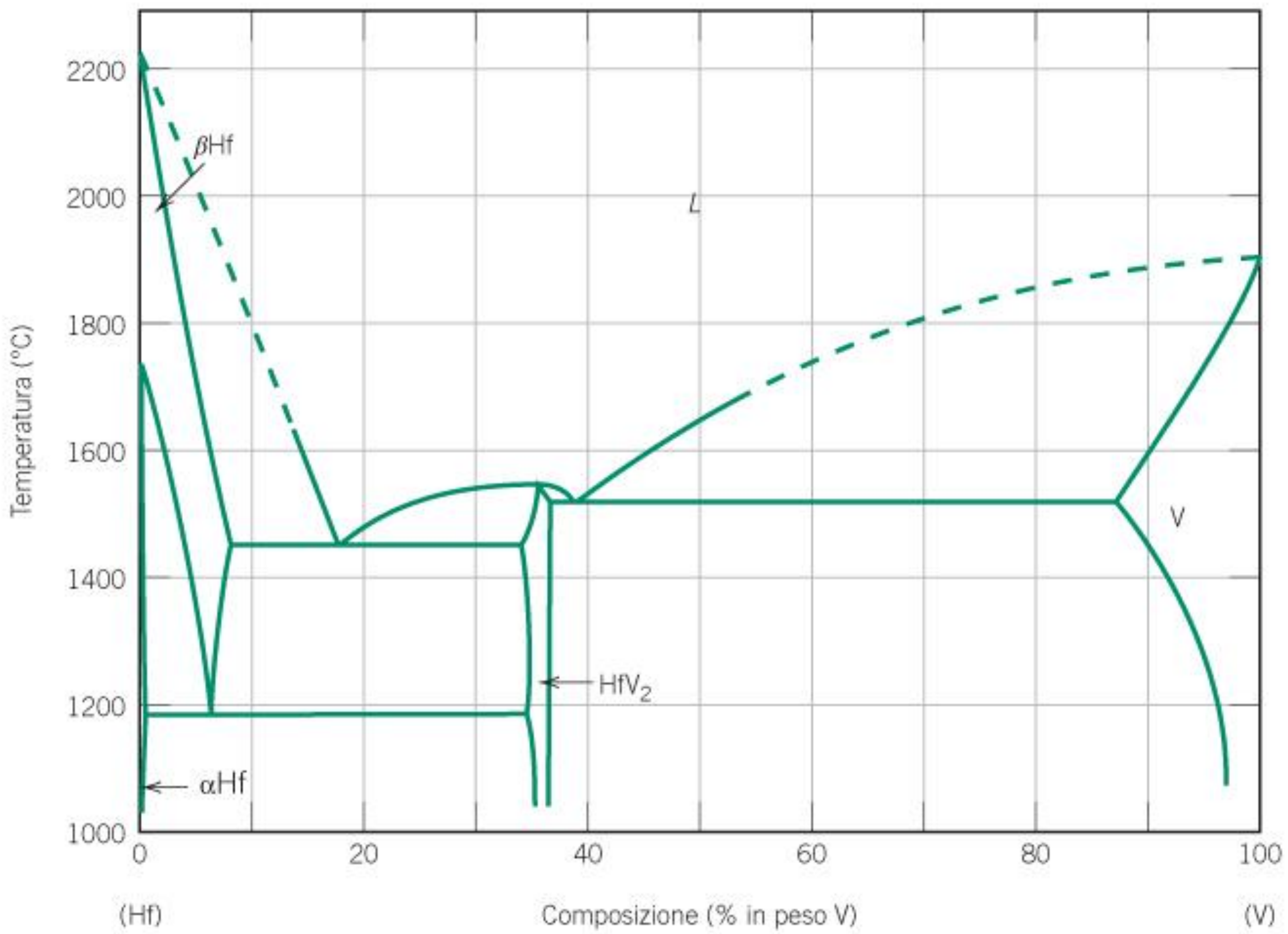
Composition of Liquid: 58%Ti

Relative amounts

% γ= (58-50)/(58-45)*100= 61.5%

%L= (50-45)/(58-45)*100=38.5%

Complete the following diagram



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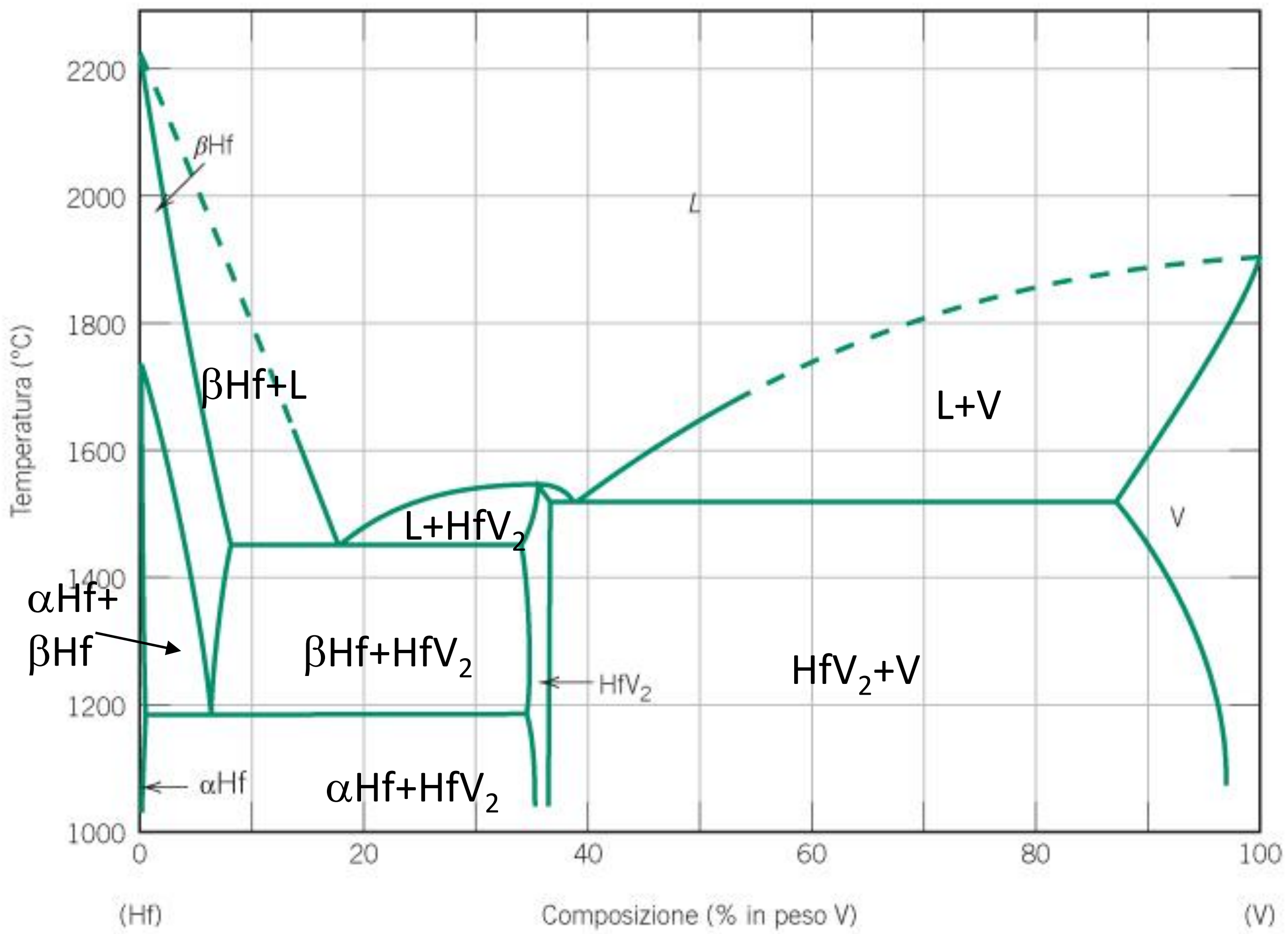

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Complete the following diagram

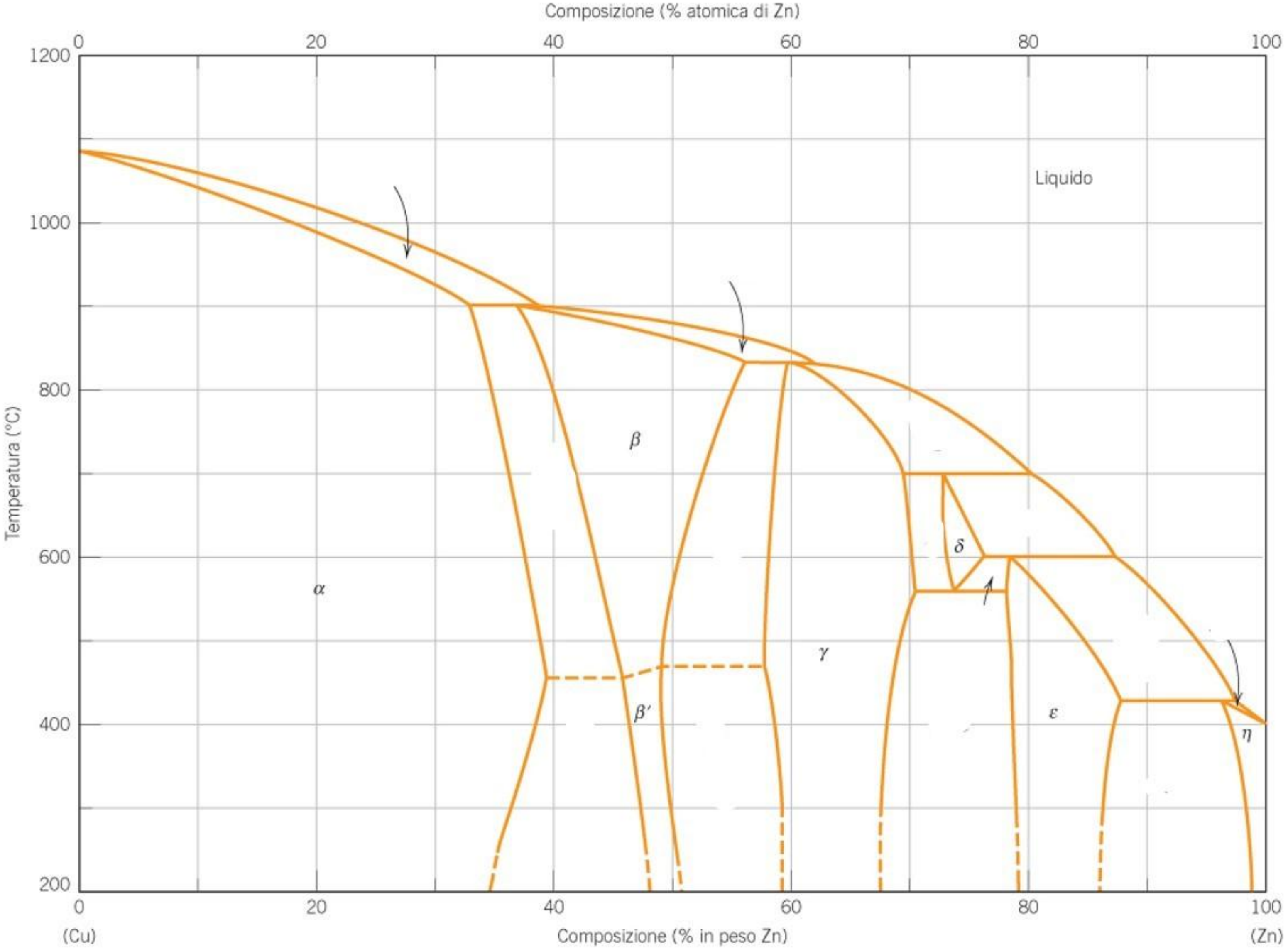

Figura 9.19 Diagramma di fase rame–zinco.

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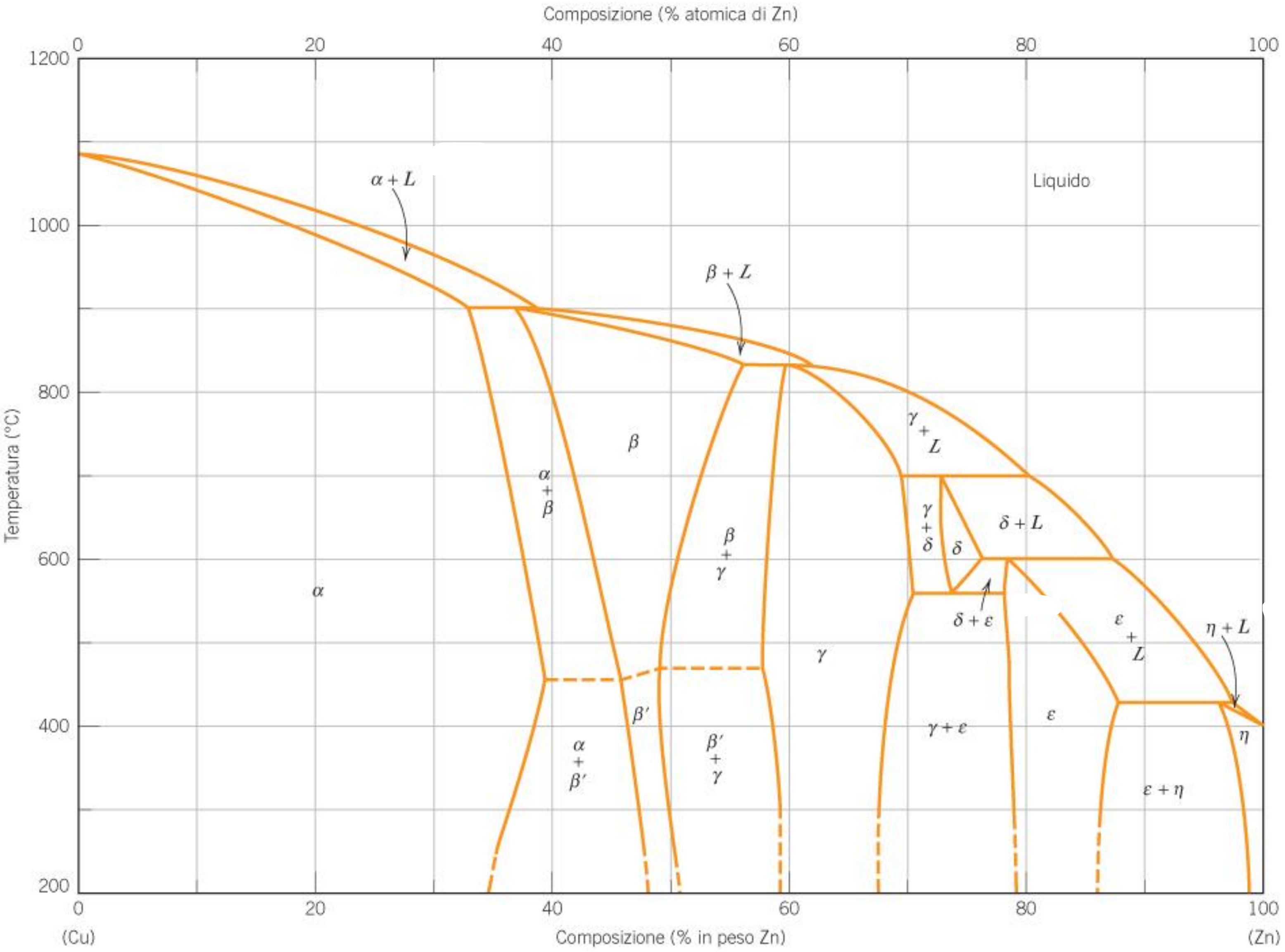

Figura 9.19 Diagramma di fase rame–zinco.

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